

Introduction to Machine Learning



Mahesh Mohan M R
Department of AI
Indian Institute of Technology Kharagpur

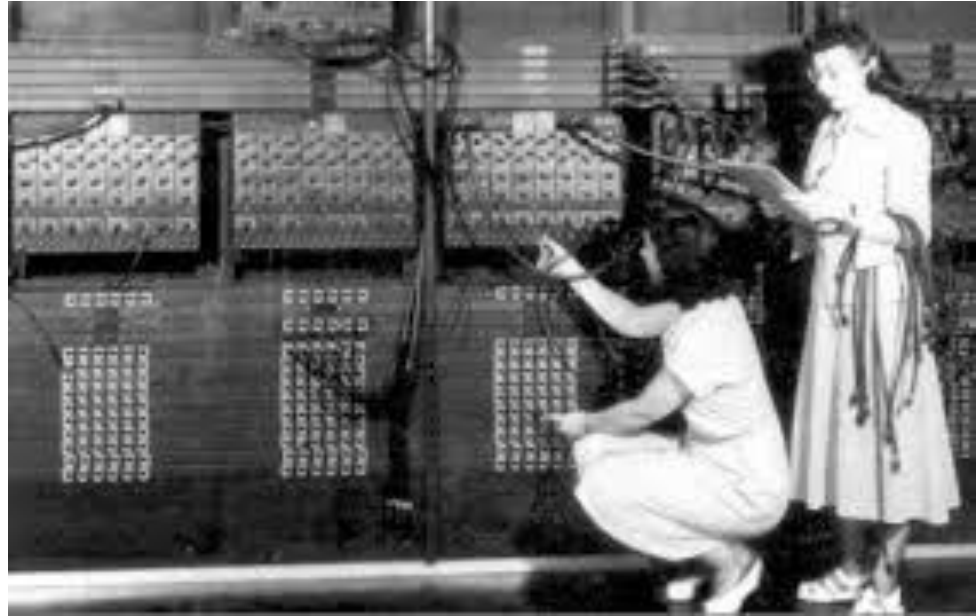
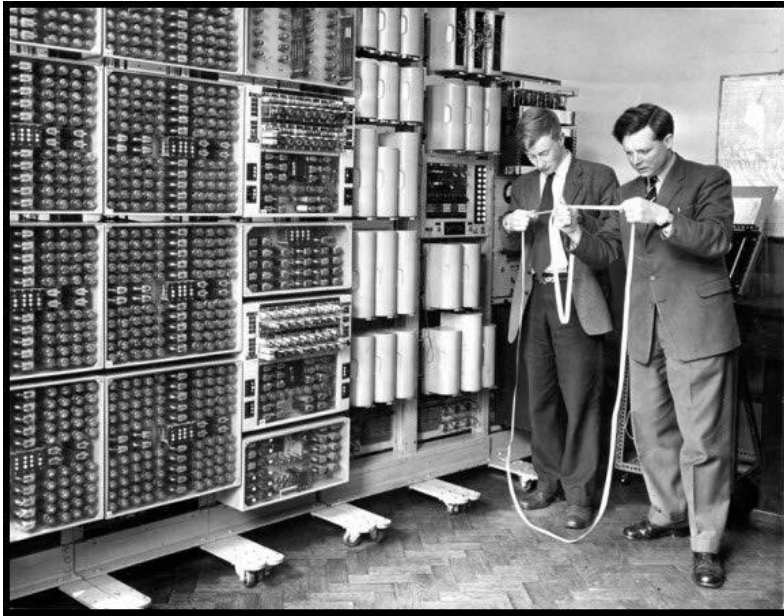
Why ML didn't take-off in 90's?

Most techniques for ML are already developed before 1990s!

Why ML didn't take-off in 90's?

Most techniques for AI are already developed before 1990s!

1. *Limited big data availability*
2. *Limited computational power to crunch data*



Why ML is trending now?

Big data availability



One trillion images.



350 million images uploaded per day.

You Tube 100 hrs of video uploaded per minute.

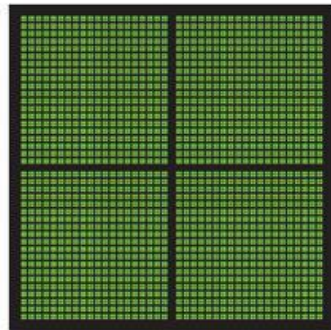


2.5 Petabytes data every minute.

Computational power to crunch data



CPU
MULTIPLE CORES

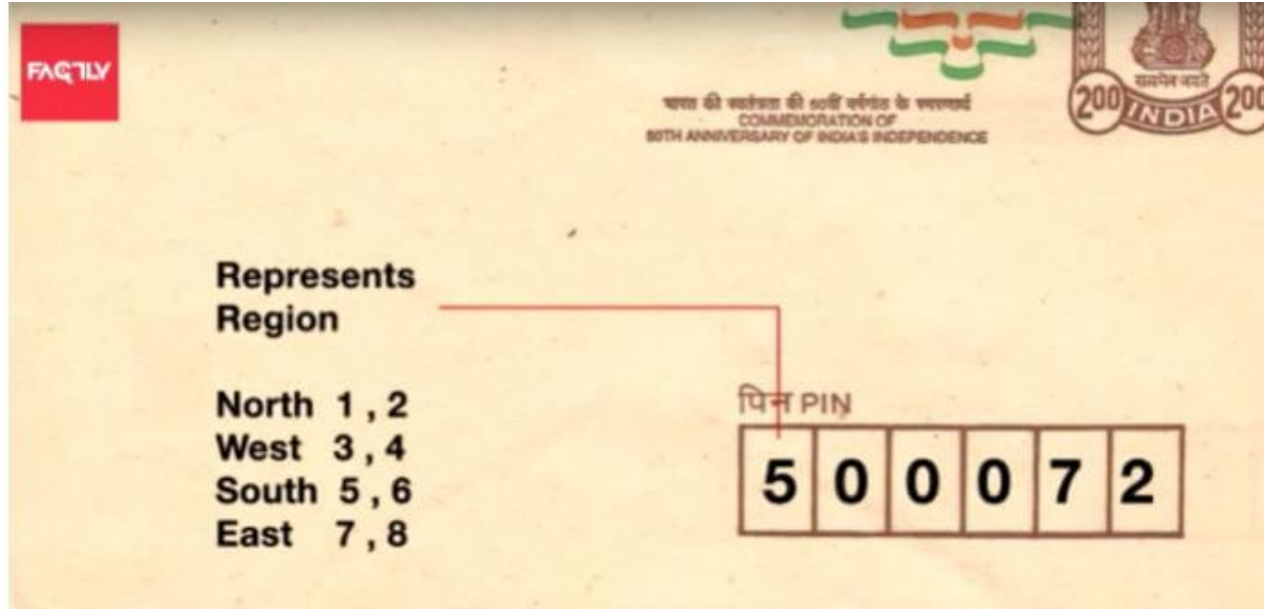


GPU
THOUSANDS OF CORES



Parallel processing units - GPUs

A Case Study: Reading Postcode



0

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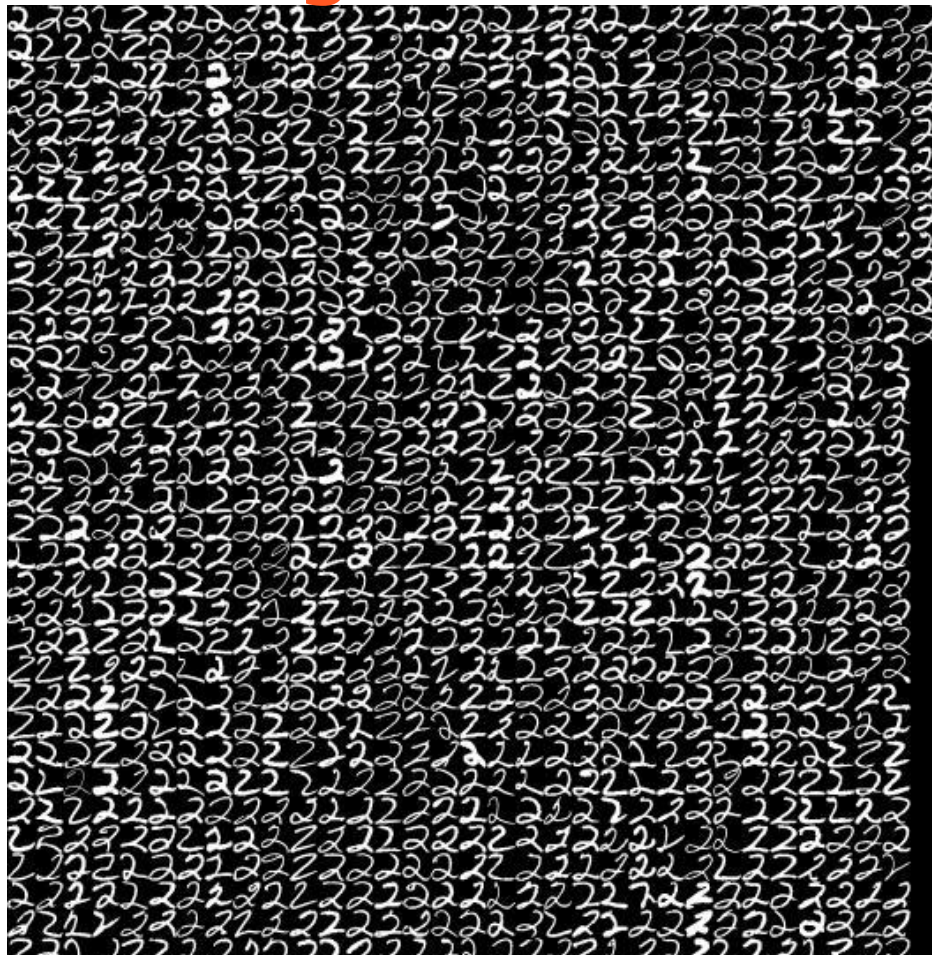
9

A Case Study: Reading Postcode



A Case Study: Reading Postcode

2



A Case Study: Reading Postcode

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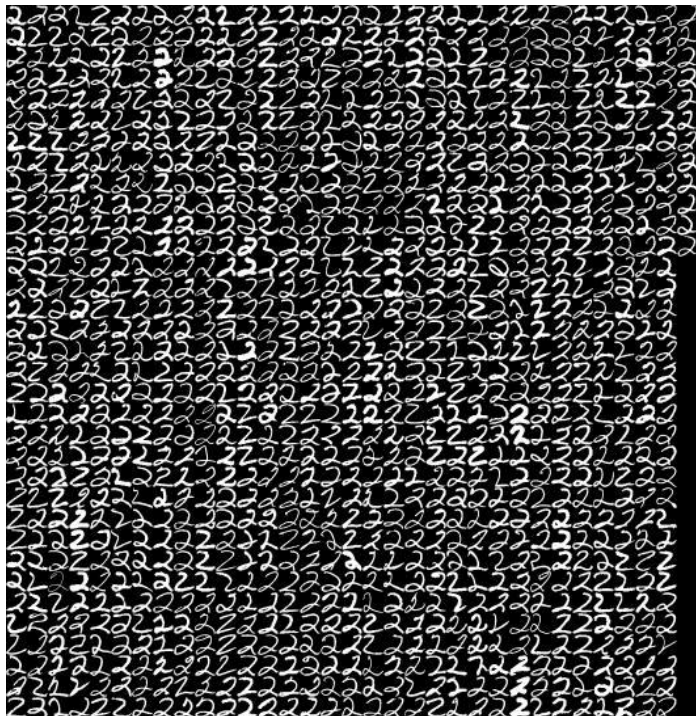
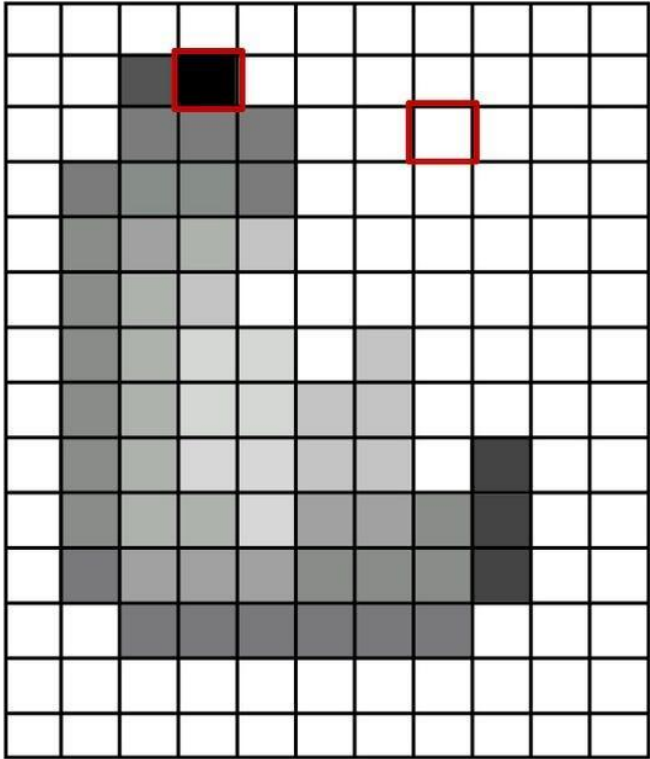
[illegible]

Image Representation in Computer

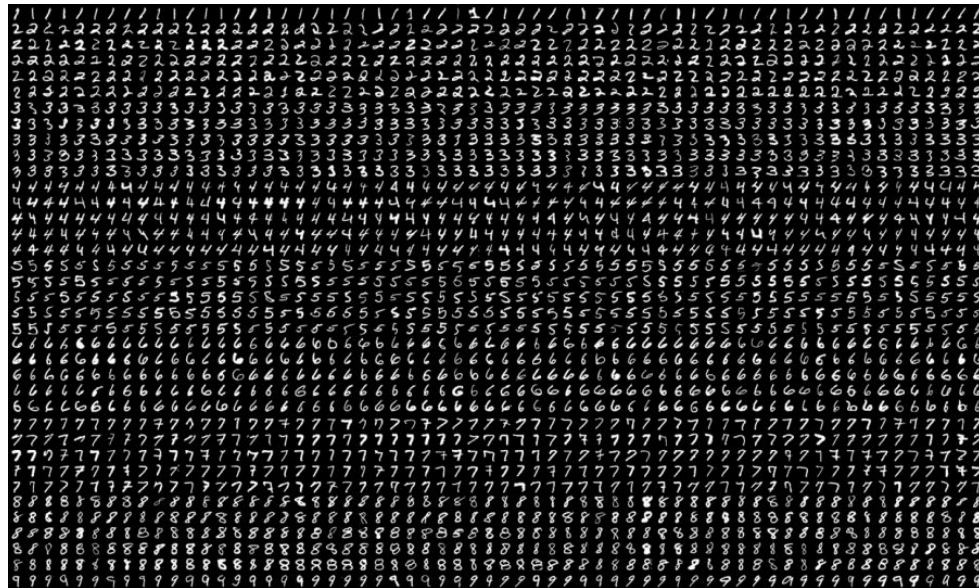
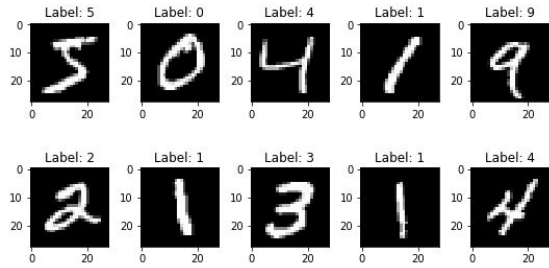


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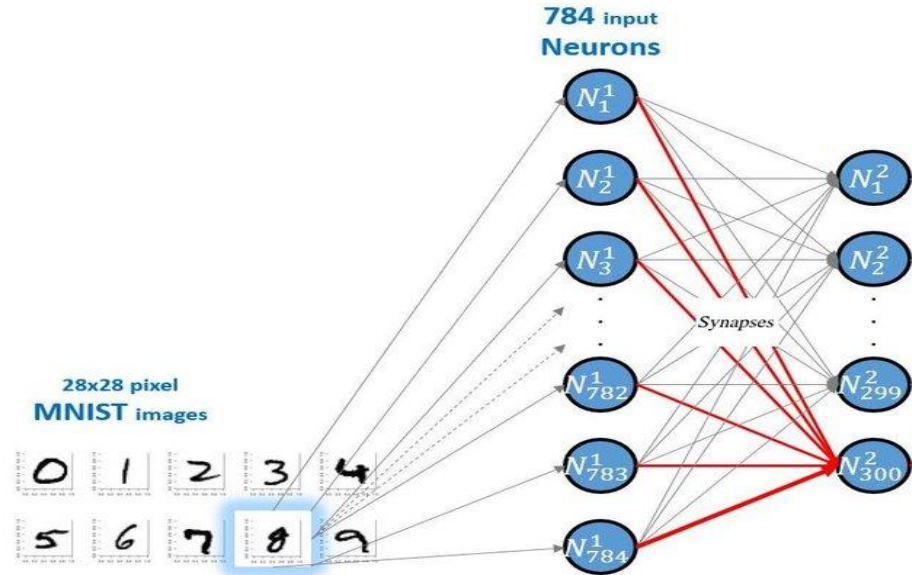
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255	255	20	0	255	255	255	255	255	255	255
255	255	75	75	255	255	255	255	255	255	255
255	75	95	95	75	255	255	255	255	255	255
255	96	127	145	175	255	255	255	255	255	255
255	127	145	175	175	175	255	255	255	255	255
255	127	145	200	200	175	175	95	255	255	255
255	127	145	200	200	175	175	95	47	255	255
255	127	145	145	175	127	127	95	47	255	255
255	74	127	127	127	95	95	95	47	255	255
255	255	74	74	74	74	74	74	255	255	255
255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255

0 = black; 255 = white

AI Based Solution: Reading Postcode

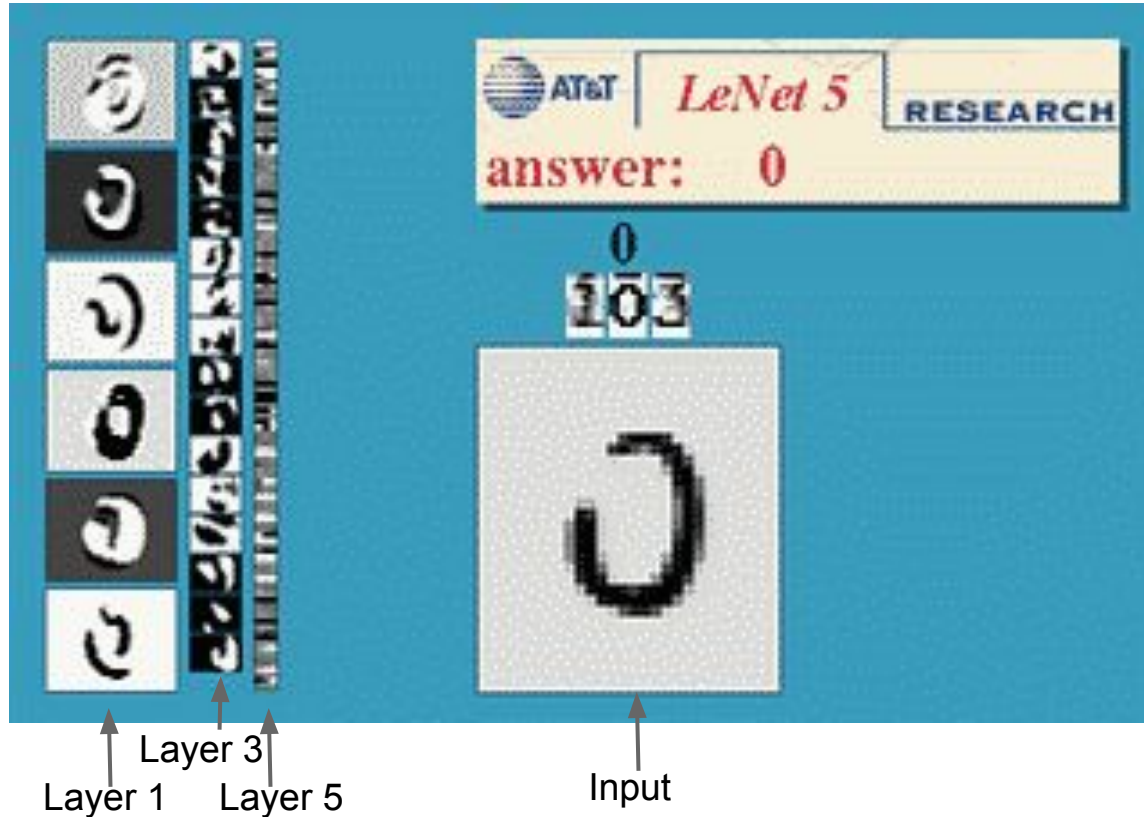


Data and Labels



Neural Network and Training

AI for Reading Postcode [LeCun *et al.*].



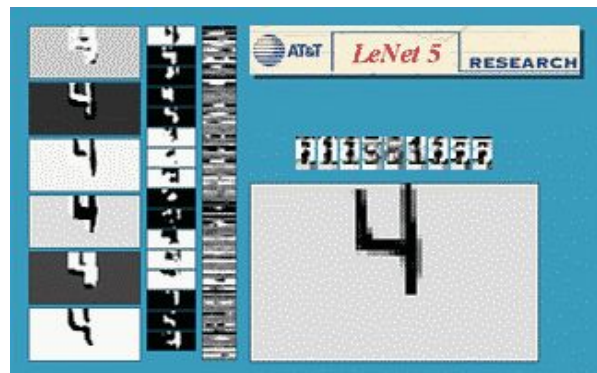
All images are 28x28 grayscale.

60k training examples.

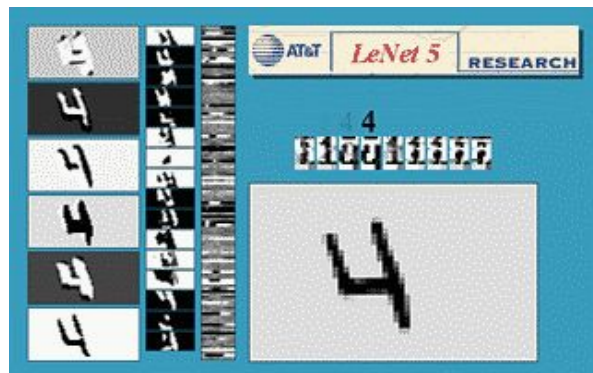
10k test examples

Output value is integer from 0-9

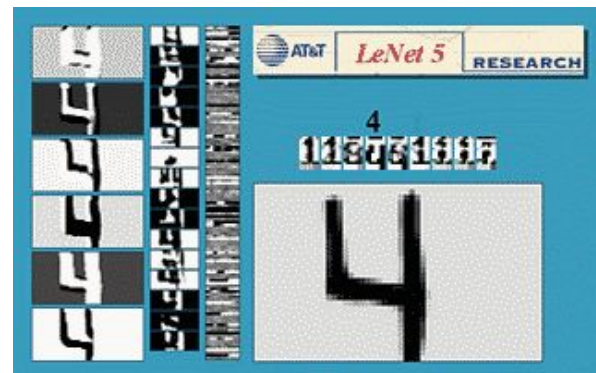
AI for Reading Postcode [LeCun *et al.*]



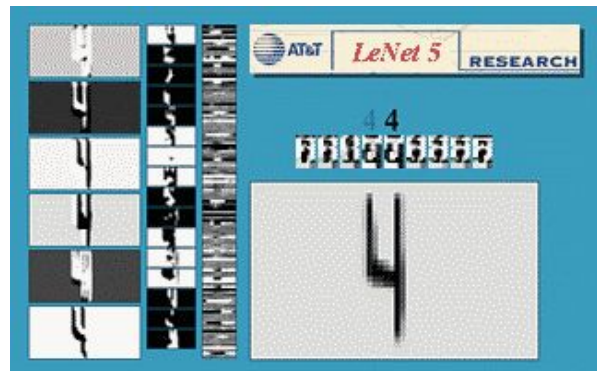
Translation invariance



Rotation invariance



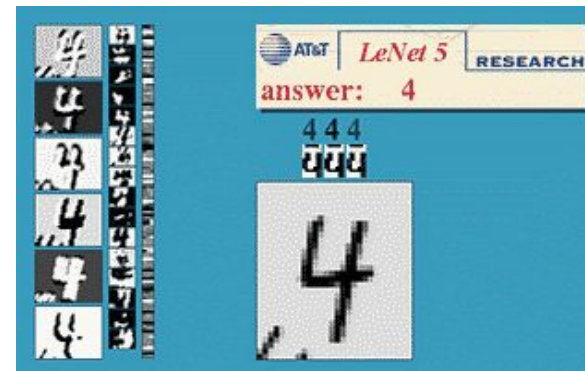
Scale invariance



Squeeze invariance

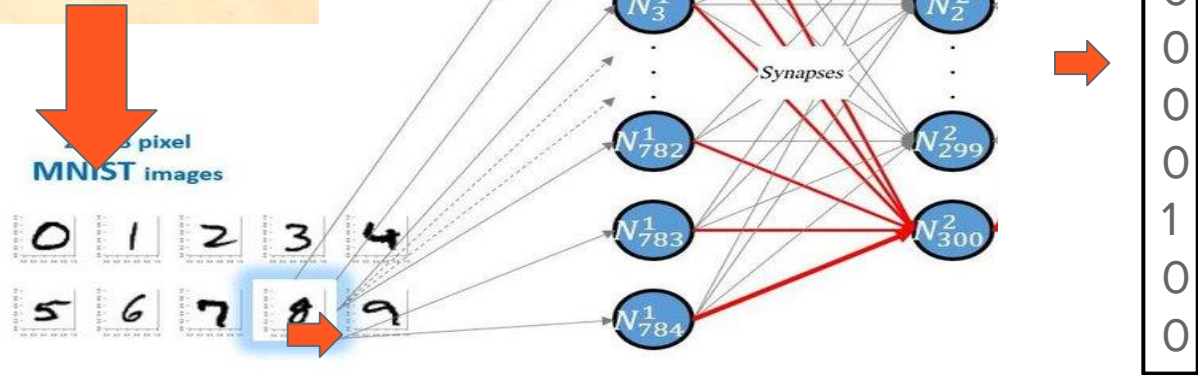
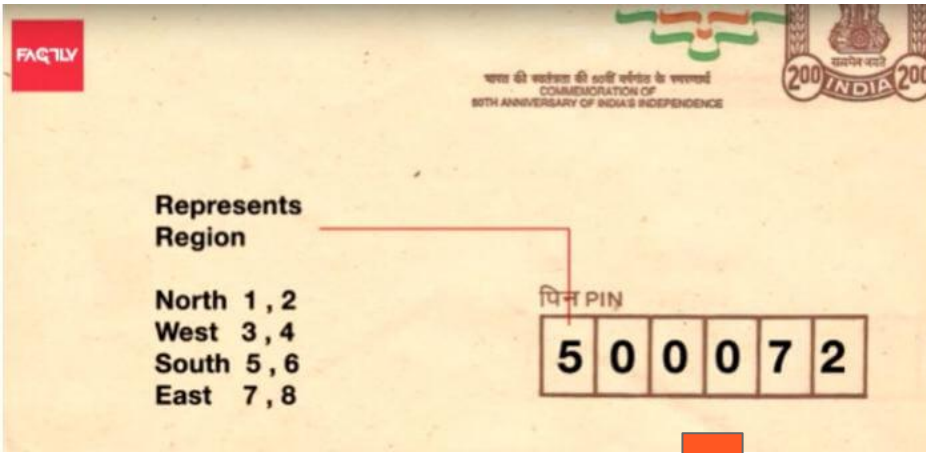


Stroke-width invariance



Noise invariance

AI: Inference/Testing

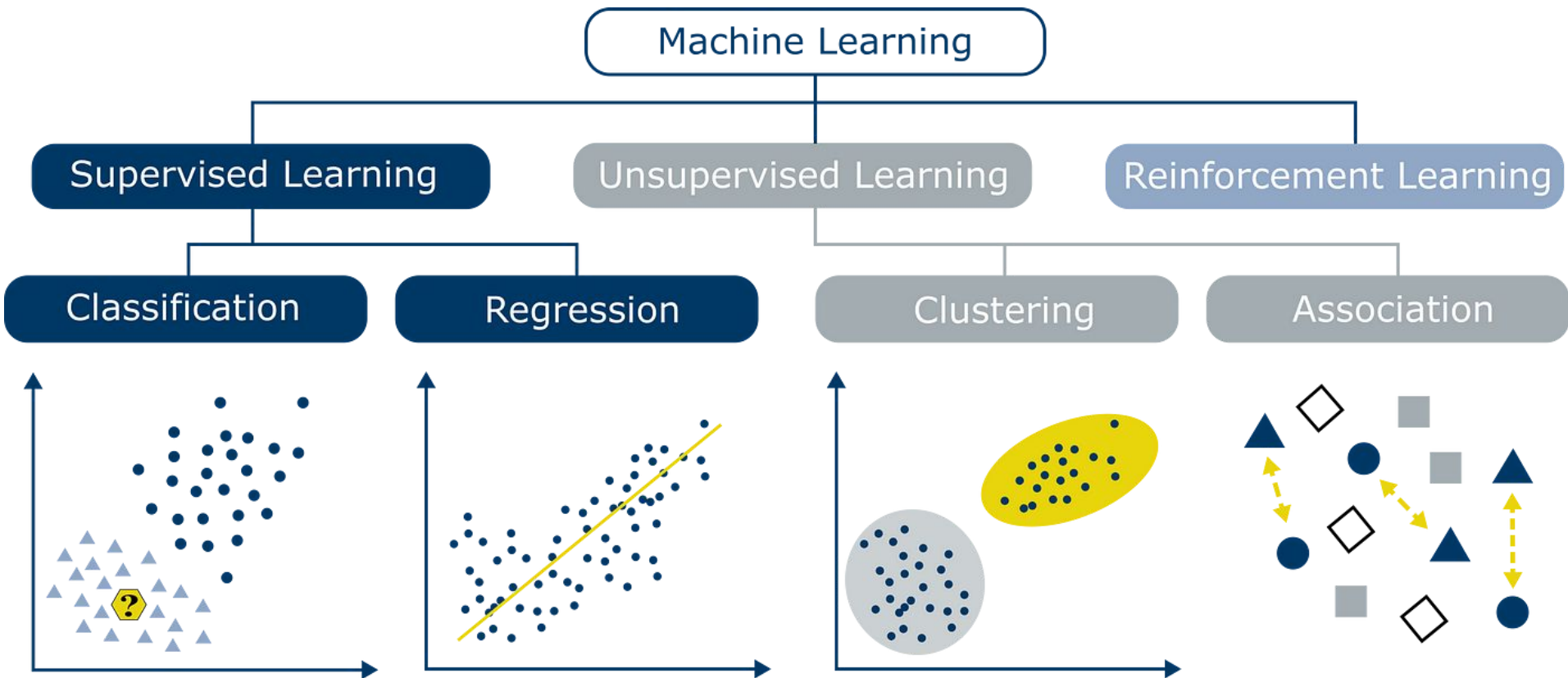


Testing samples extracted from envelope

Artificial Neural Network (ANN)

O/p

Outline

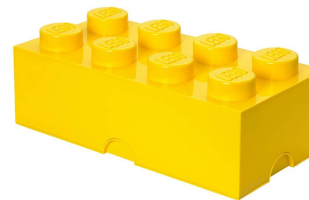


Introduction to Supervised Learning

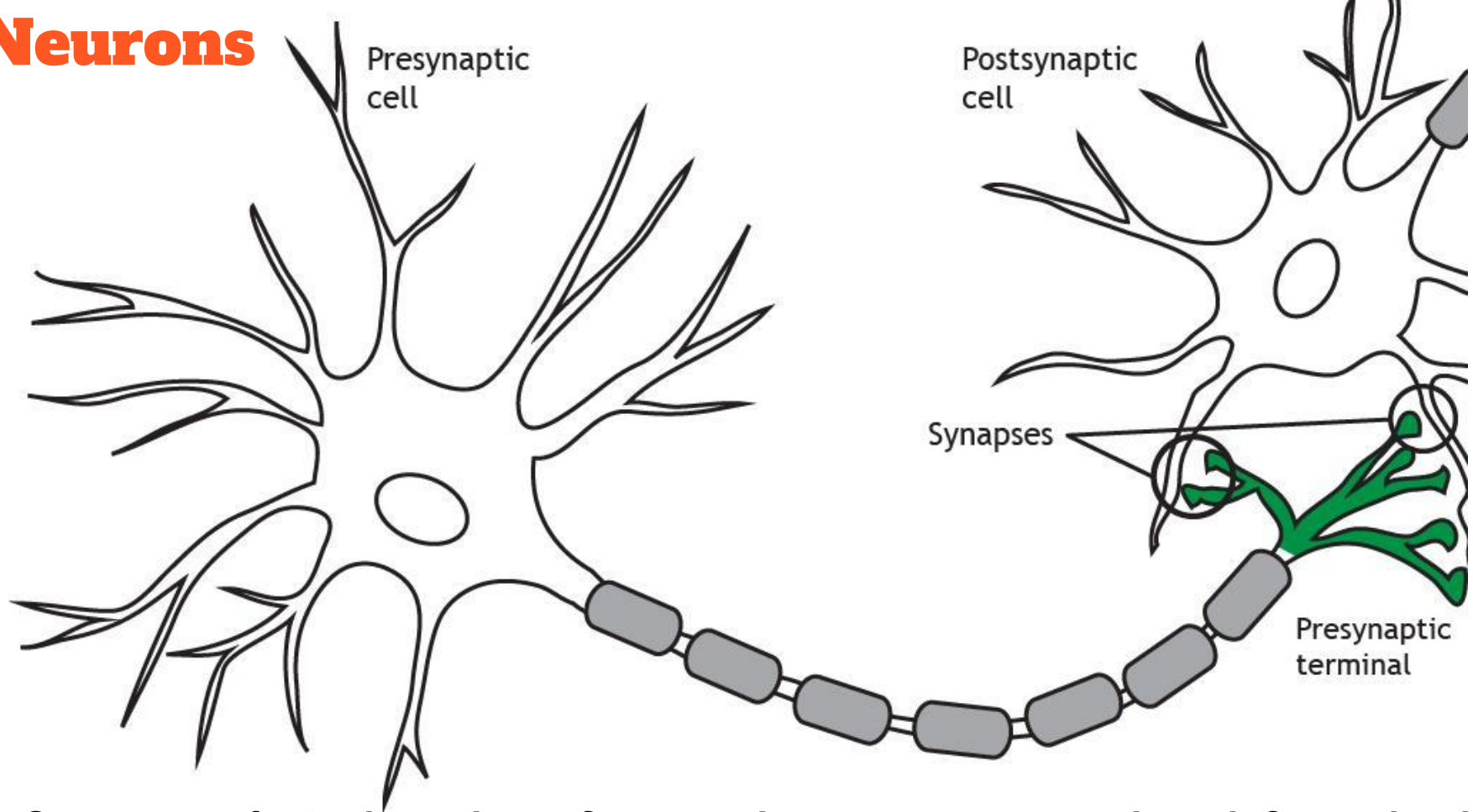




Yellow color



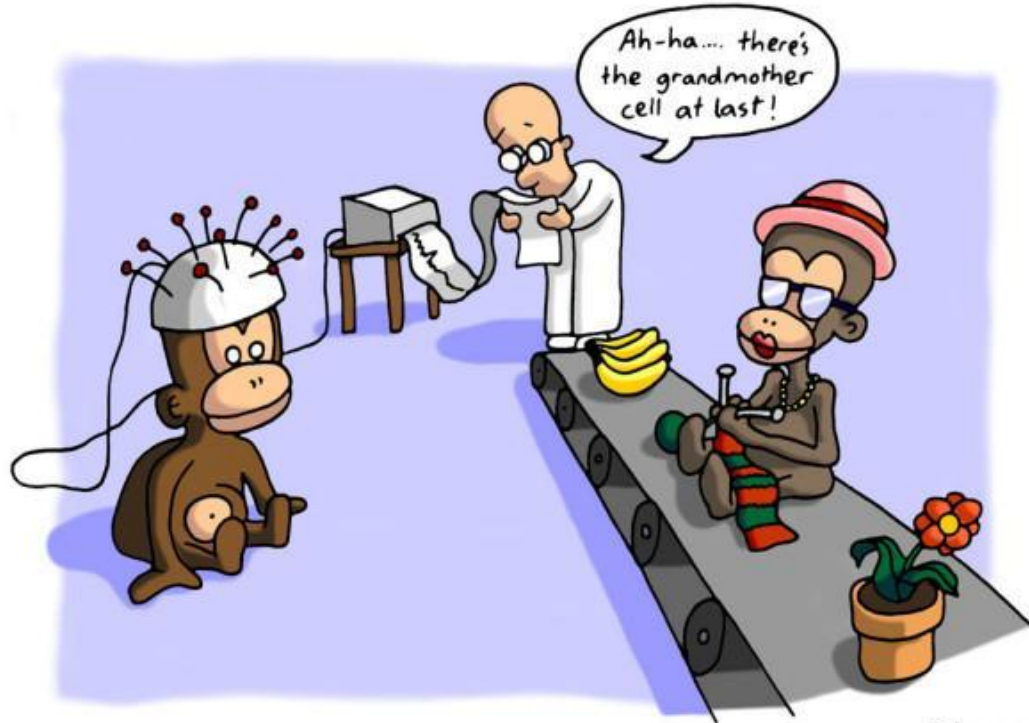
Neurons



Synapses refer to **the points of contact between neurons where information is passed from one neuron to the next.**

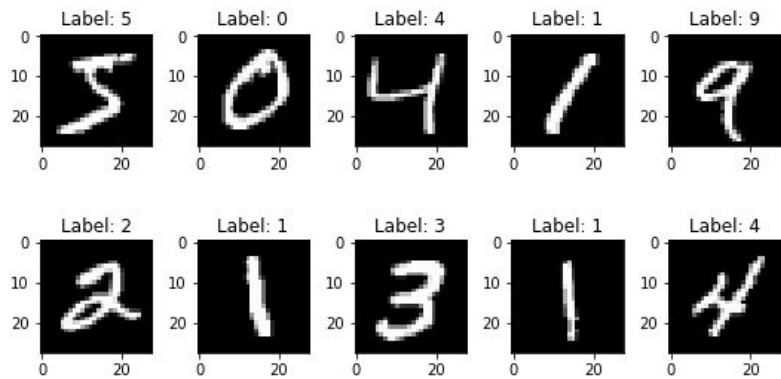
Biological Motivation - Grandmother Cell

The grandmother cell is a hypothetical neuron that represents a complex but specific concept or object proposed by cognitive scientist Jerry Letvin in 1969.

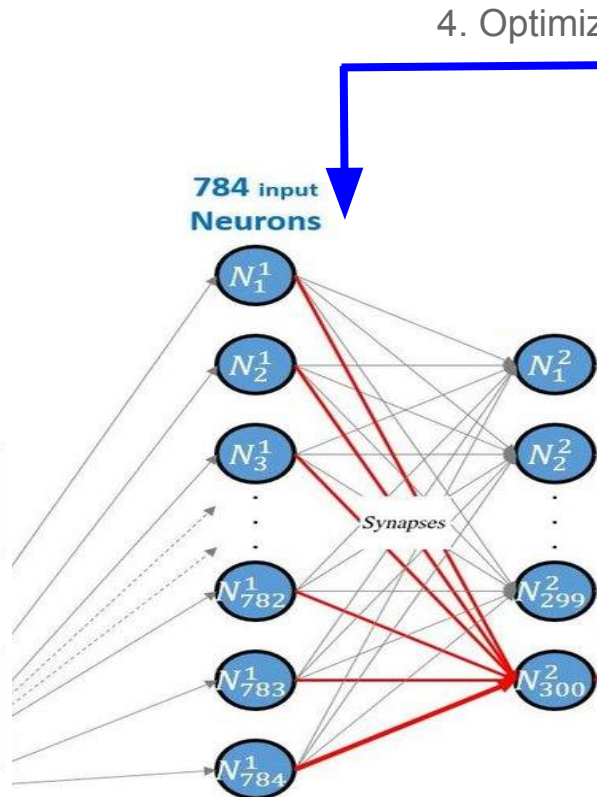


AI: How to Solve it?

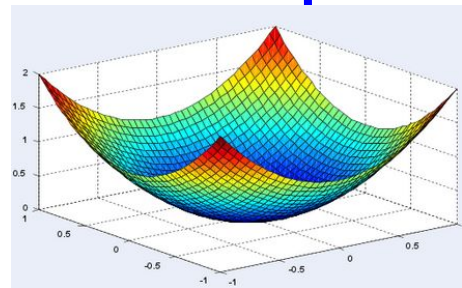
1. Collect Labelled Dataset
2. Design Machine Learning Model
3. Define Loss Function
4. Optimize weights



1. Collect Labelled dataset



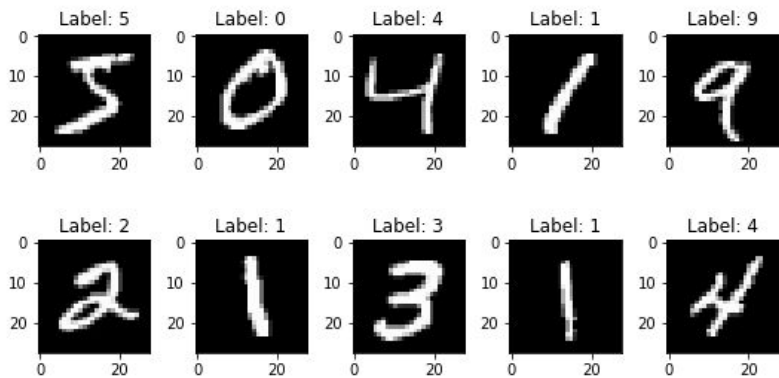
2. Design Artificial Neural Network



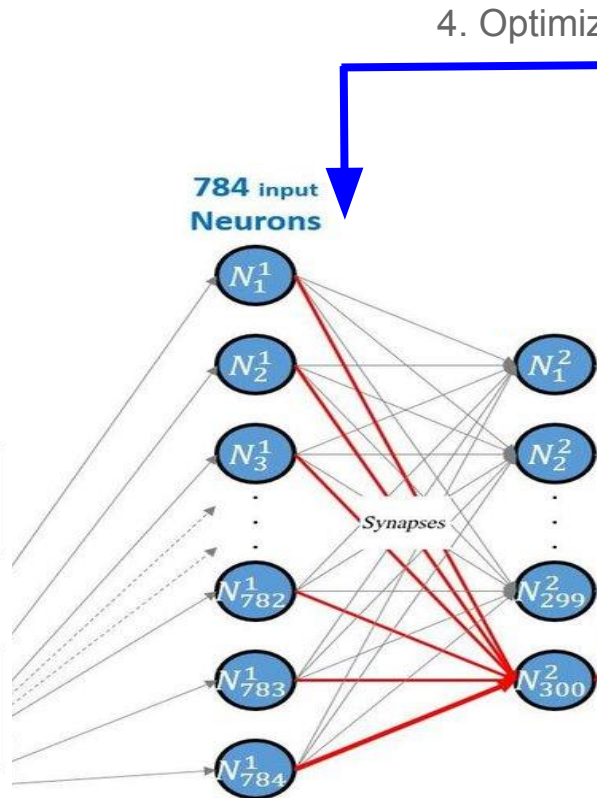
3. Loss Function

AI: How to Solve it?

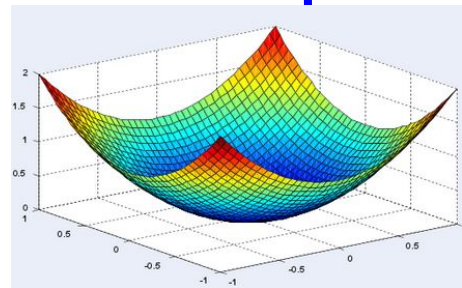
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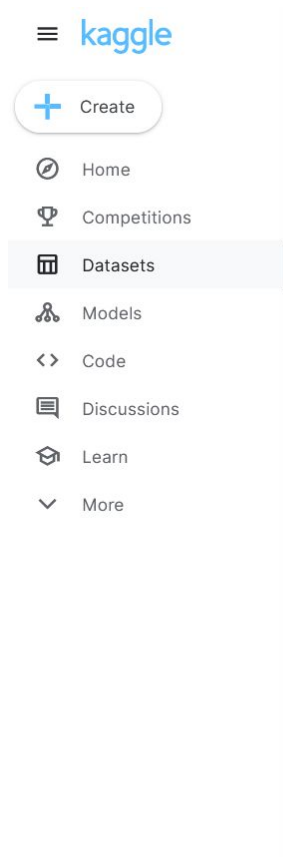
2. Design Artificial Neural Network



3. Loss Function

4. Optimize Weight

Collect Labelled Dataset



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Education

Classification

Computer Vision

NLP

Data Visualization

Pre-Trained Model



American House Prices

Jeremy Larcher · Updated 2 days ago

Usability **9.4** · 682 kB

1 File (CSV)



Books Sales and Ratings

The Devastator · Updated 5 days ago

Usability **9.4** · 55 kB

1 File (CSV)



Breakfast Sales and Customer Engagement Data

Prajwal Dongre · Updated 12 days ago

Usability **10.0** · 5 kB

1 File (CSV)



Nurse Stress Prediction Wearable Sensors

Priyank Raval · Updated 6 days ago

Usability **9.4** · 80 MB

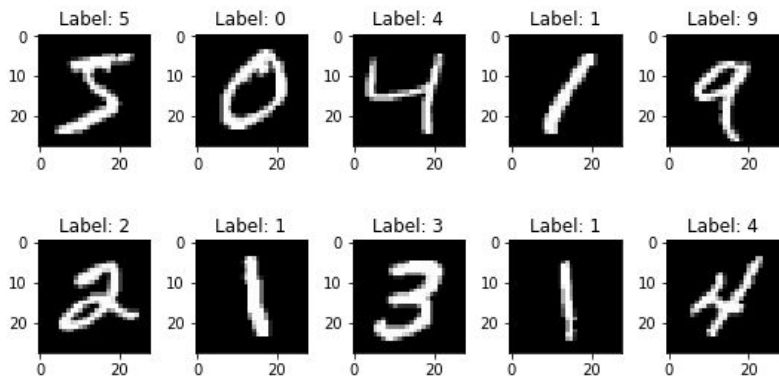
1 File (CSV)

See All

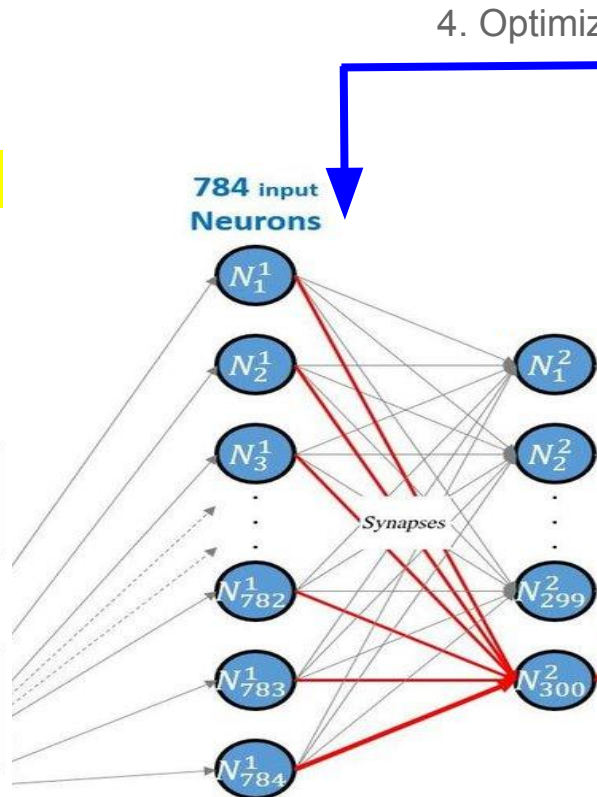
A stylized illustration of a person with dark, curly hair, wearing a white lab coat, interacting with several floating digital screens. The screens display various data visualizations, including a line graph, a bar chart, and a pie chart. The person is pointing at one of the screens with their right hand.

AI: How to Solve it?

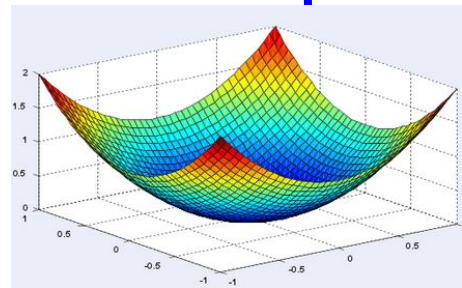
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1. Collect Labelled dataset



2. Design Artificial Neural Network

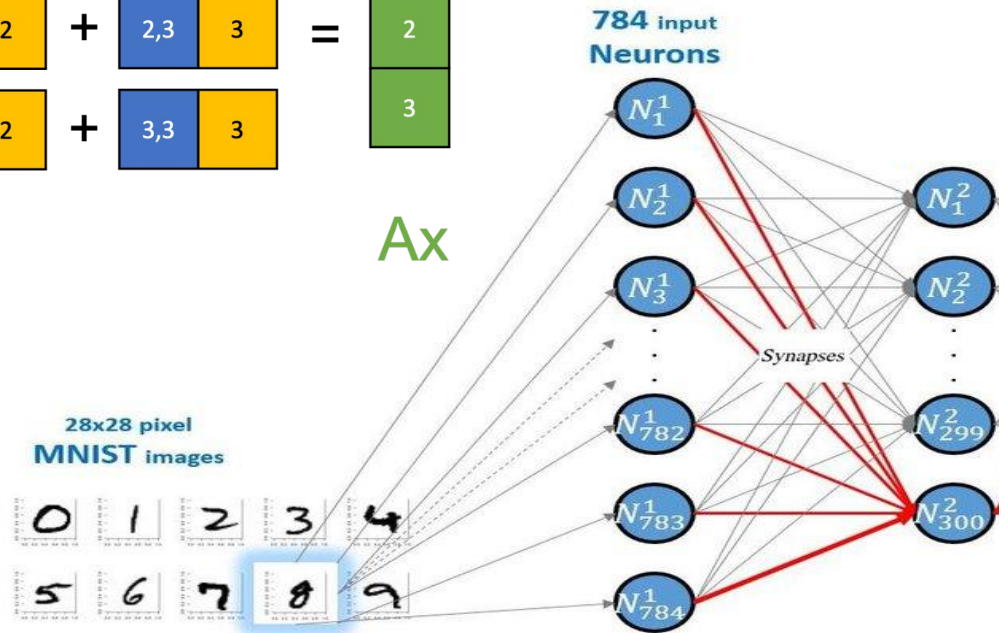


3. Loss Function

Design ANN Architecture

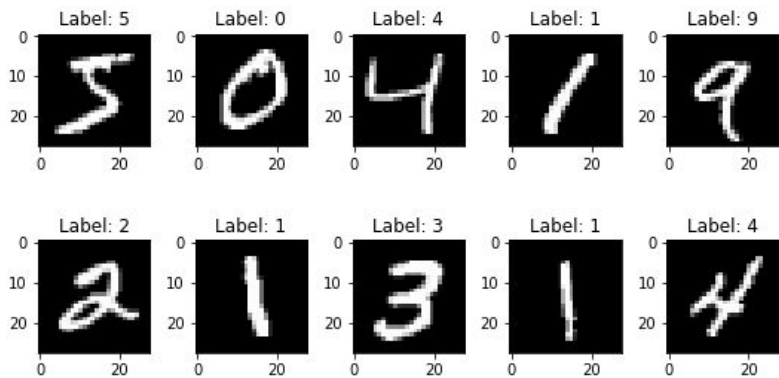
$$\begin{bmatrix} 1,1 & 1,2 & 1,3 \\ 2,1 & 2,2 & 2,3 \\ 3,1 & 3,2 & 3,3 \end{bmatrix} \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1,1 & 1 & 1,2 & 2 & 1,3 & 3 \\ 2,1 & 1 & 2,2 & 2 & 2,3 & 3 \\ 3,1 & 1 & 3,2 & 2 & 3,3 & 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

A
X
Ax

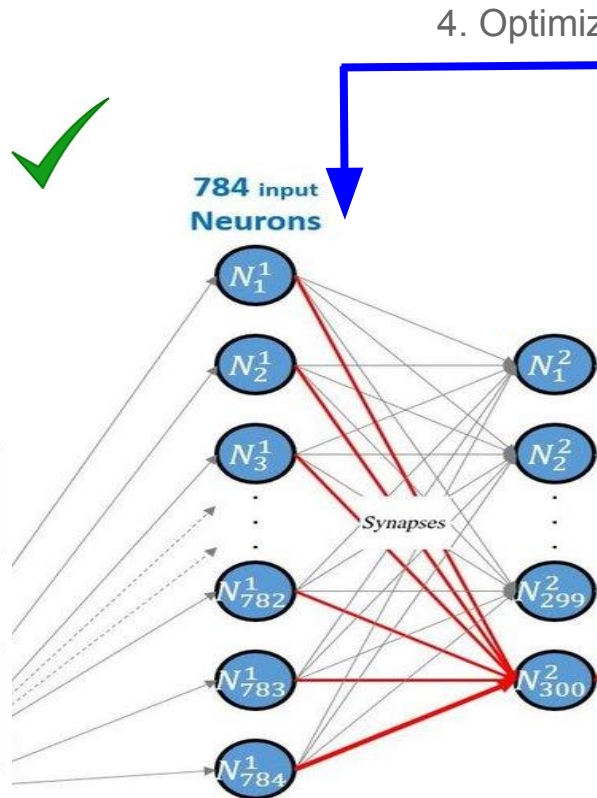


AI: How to Solve it?

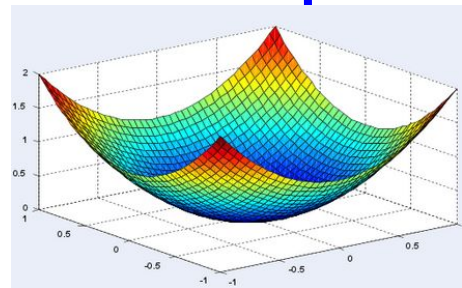
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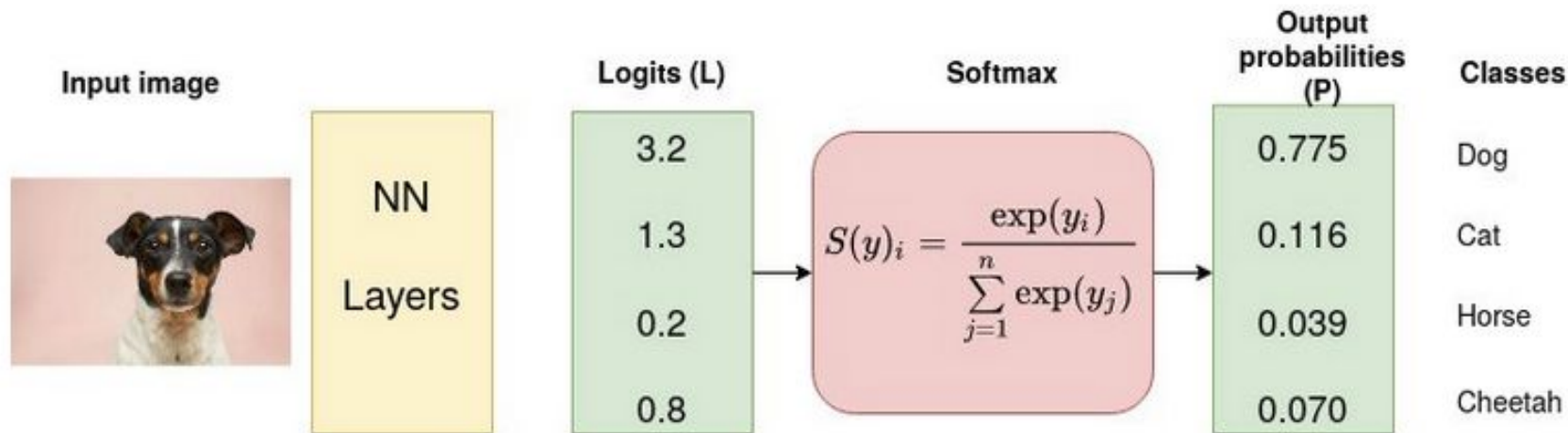


2. Design Artificial Neural Network

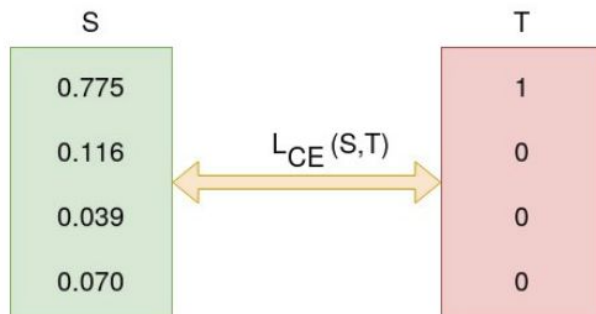


3. Loss Function

Cross-entropy Loss



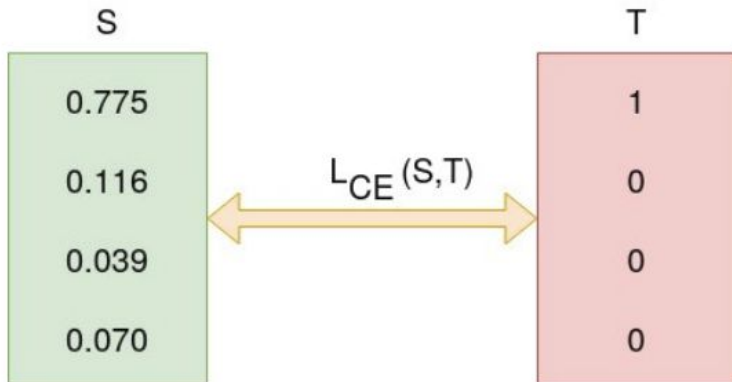
What we want?



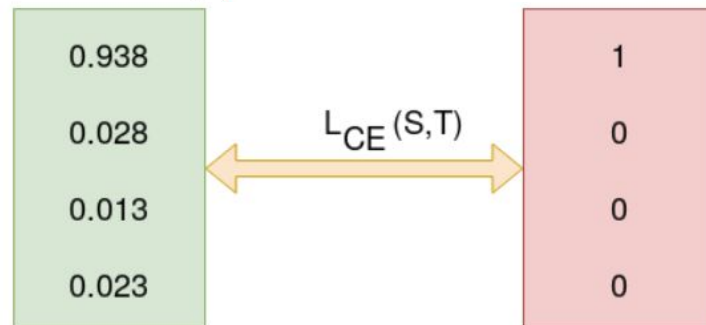
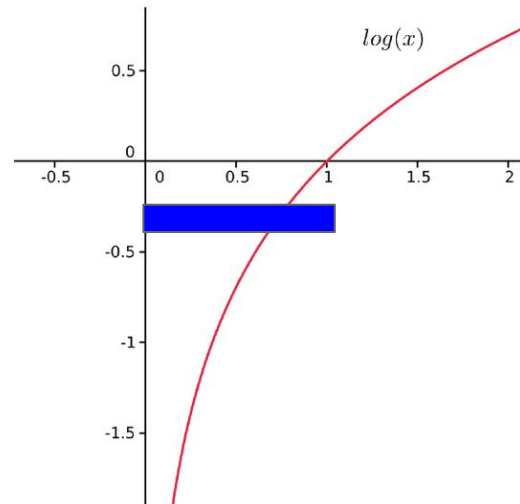
Cross-entropy Loss?

$$L_{CE} = - \sum_{i=1}^n t_i \log(p_i), \text{ for } n \text{ classes,}$$

where t_i is the truth label and p_i is the Softmax probability for the i^{th} class.

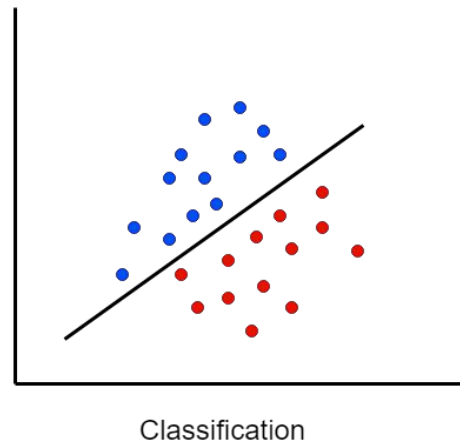
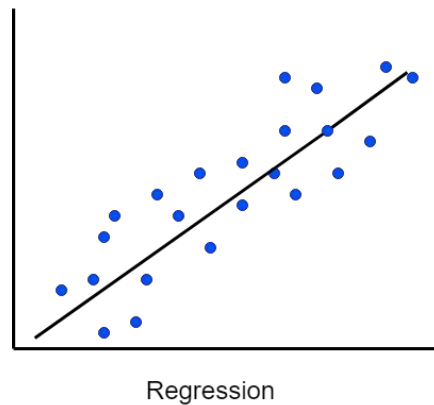
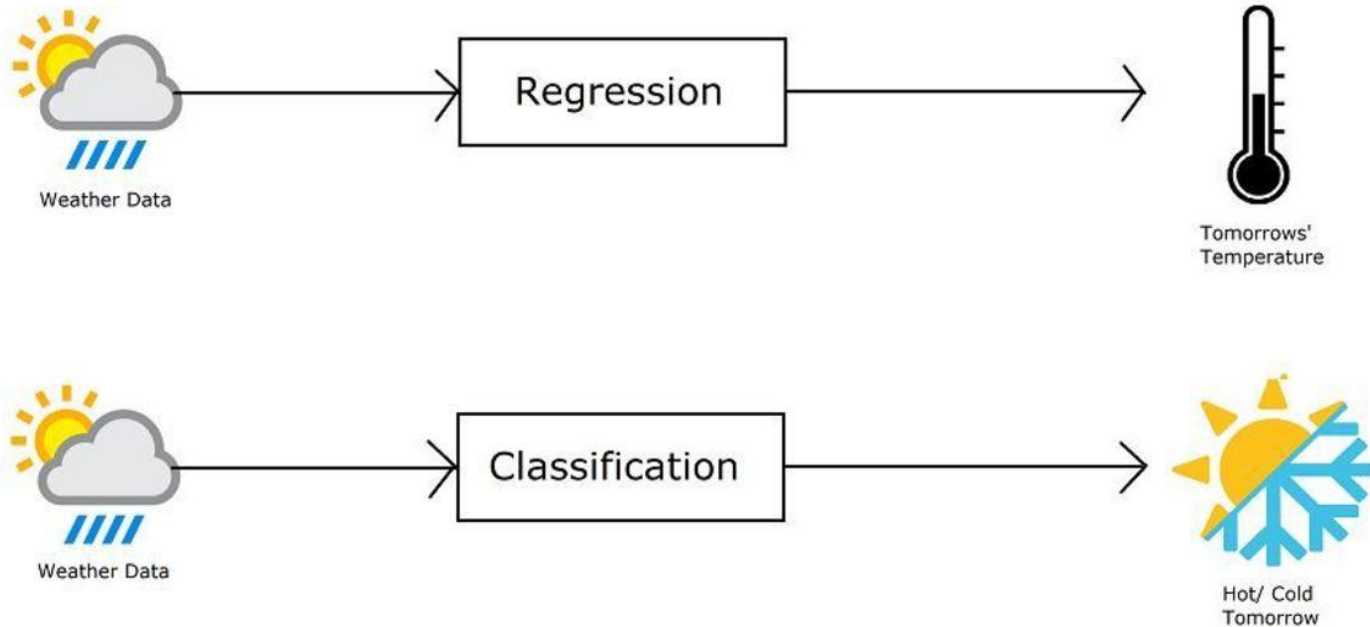


$$\begin{aligned} L_{CE} &= - \sum_{i=1} T_i \log(S_i) \\ &= - [1 \log_2(0.775) + 0 \log_2(0.126) + 0 \log_2(0.039) + 0 \log_2(0.070)] \\ &= - \log_2(0.775) \\ &= 0.3677 \end{aligned}$$

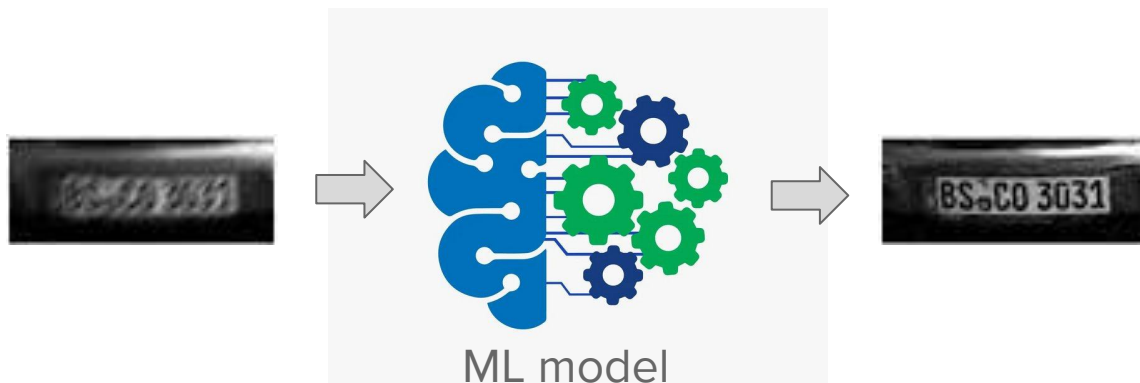


$$\begin{aligned} L_{CE} &= -1 \log_2(0.936) + 0 + 0 + 0 \\ &= 0.095 \end{aligned}$$

Classification vs Regression



A Real Life Application



Input and Output are similar

MSE Loss

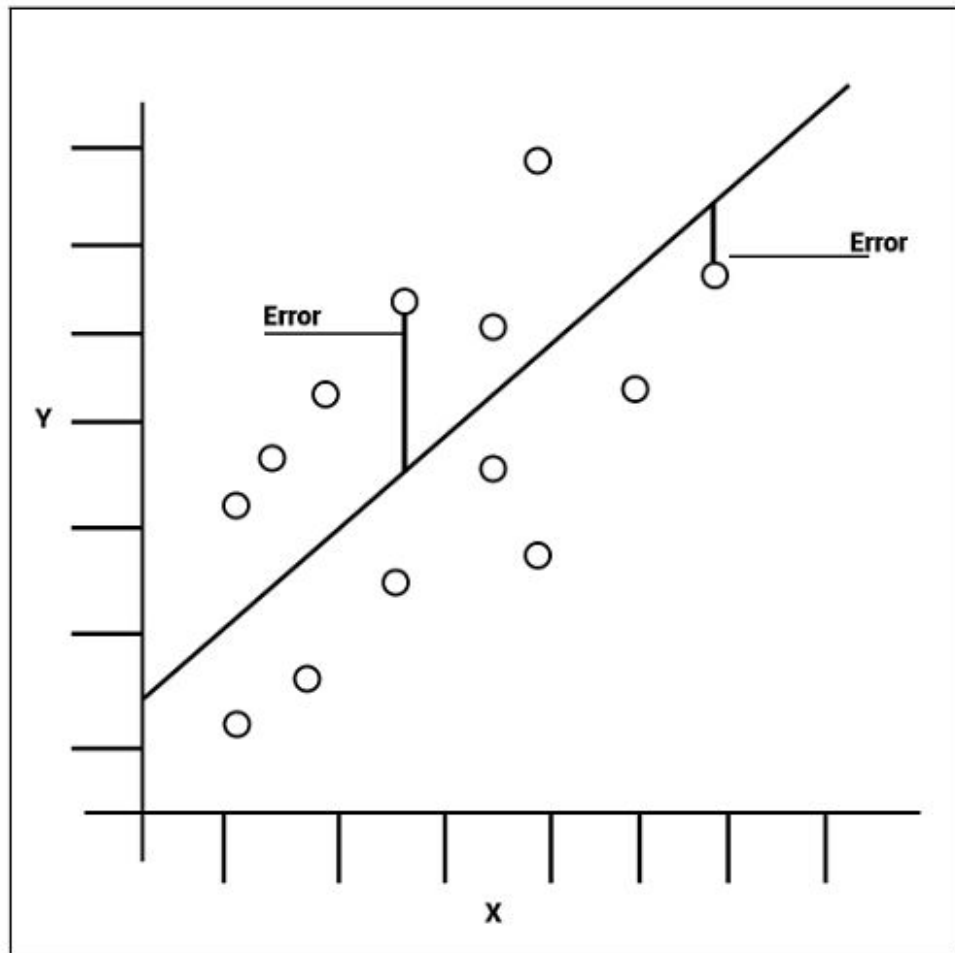
$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

MSE = mean squared error

n = number of data points

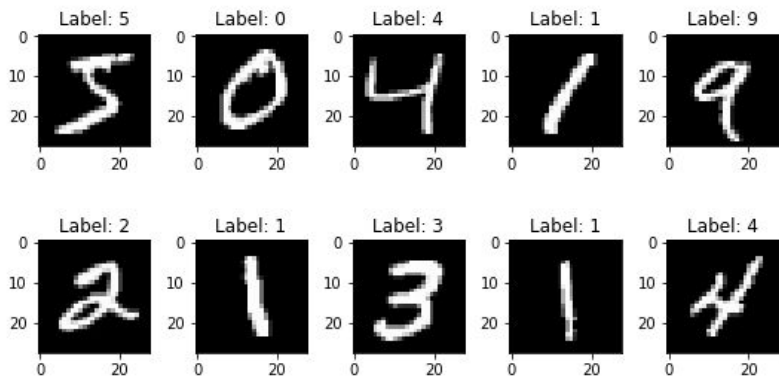
Y_i = observed values

$\hat{Y}_i$ = predicted values

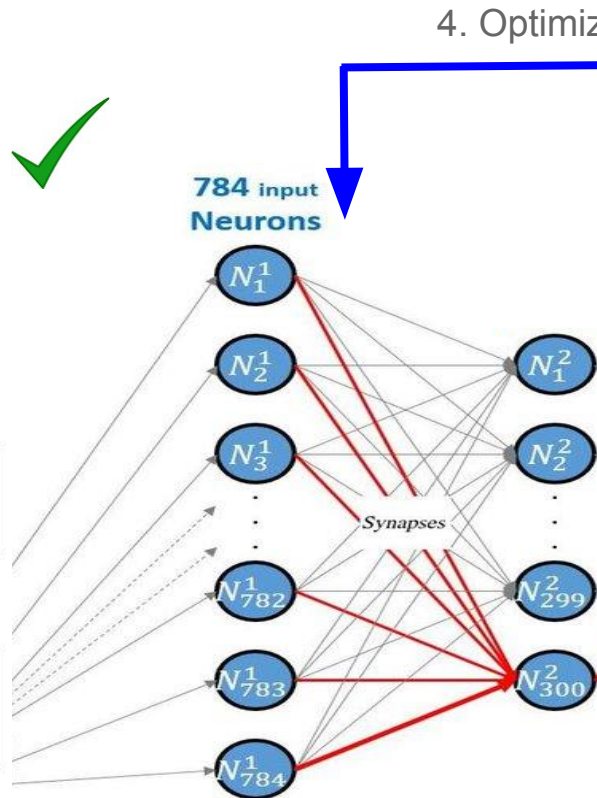


AI: How to Solve it?

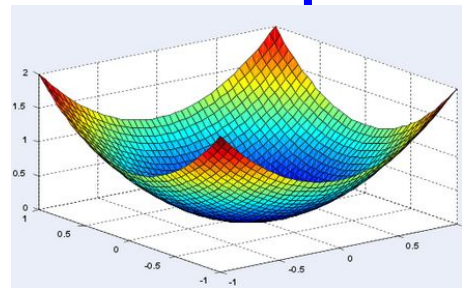
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1. Collect Labelled dataset



2. Design Artificial Neural Network

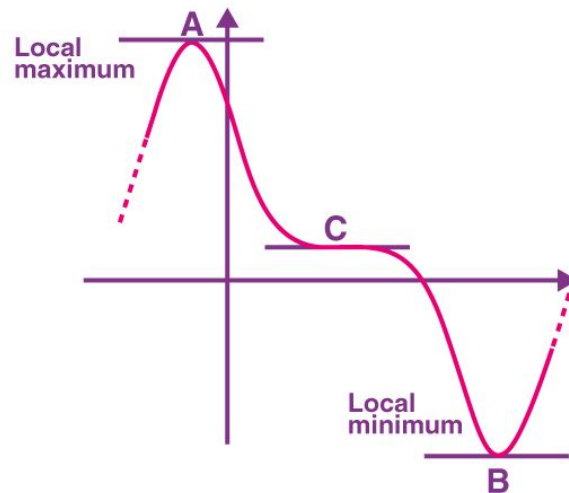
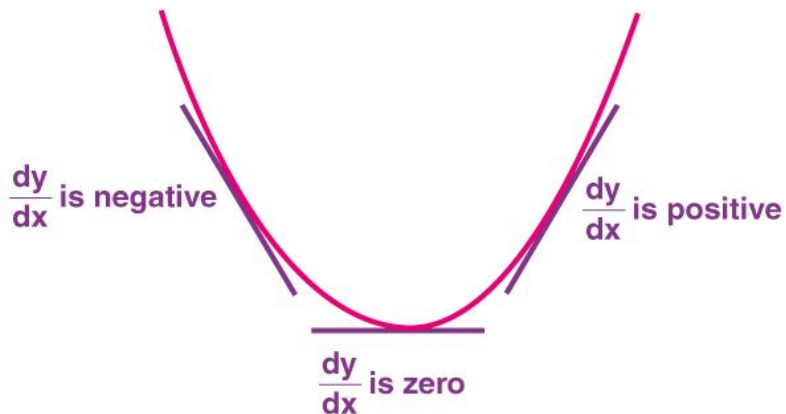


3. Loss Function

4. Optimize Weight

Optimizing ANN: Closed Form Expression

	Maximum	Minimum
Necessary condition	$\frac{dy}{dx} = 0$	$\frac{dy}{dx} = 0$
Sufficient condition	$\frac{dy}{dx} = 0 ; \frac{d^2y}{dx^2} < 0$	$\frac{dy}{dx} = 0 ; \frac{d^2y}{dx^2} > 0$

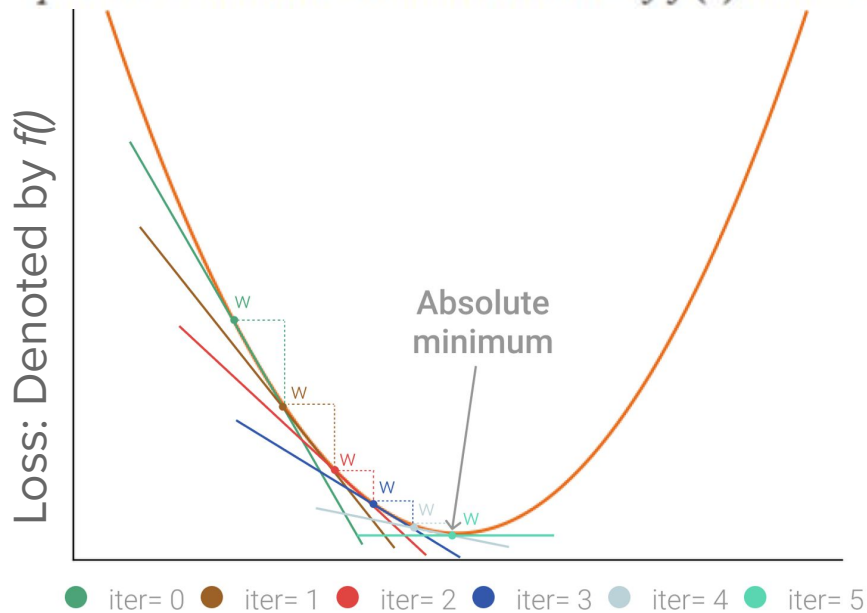


Optimizing ANN: Gradient Descent

Definition 1 Suppose f is a real valued function and a is a point in its domain of definition. The derivative of f at a is defined by

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

provided this limit exists. Derivative of $f(x)$ at a is denoted by $f'(a)$.



ANN weight: denoted by a

What happens to Loss $f(a+h) - f(a)$ if the weight update h is $-\lambda f'(a)$?

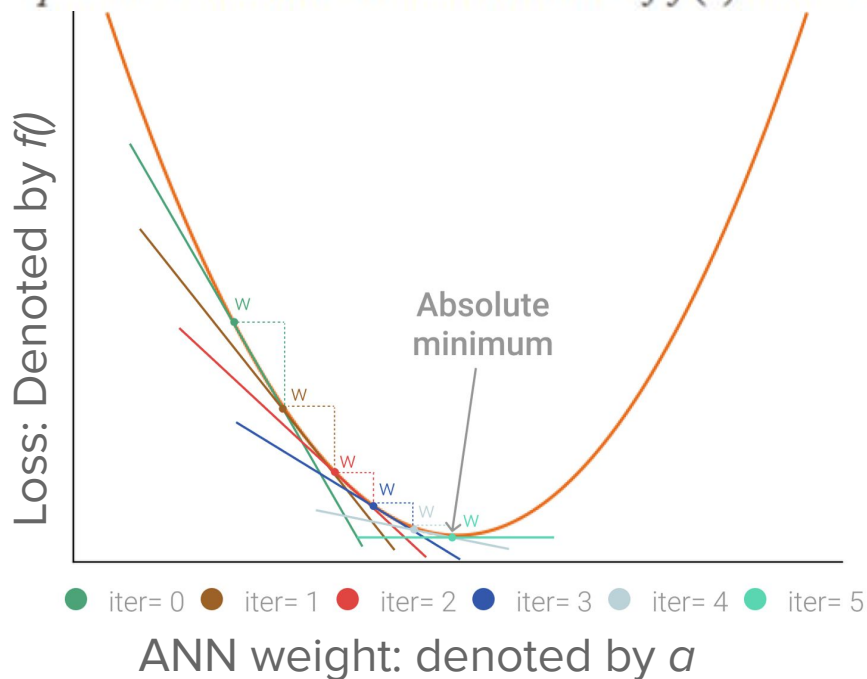
Note: $\lambda > 0$

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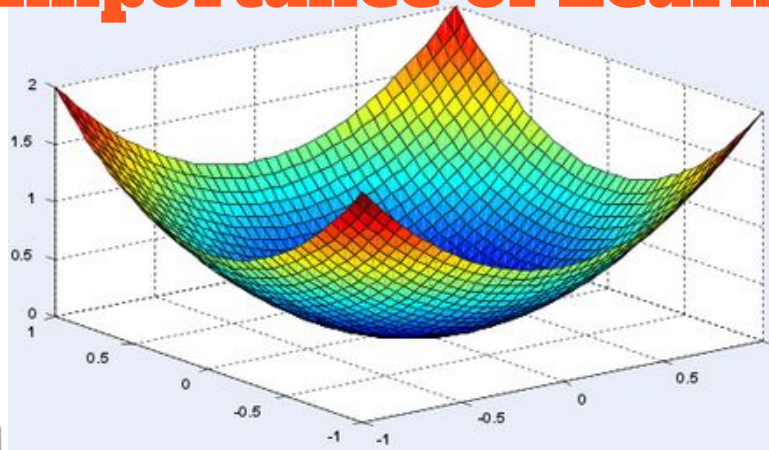
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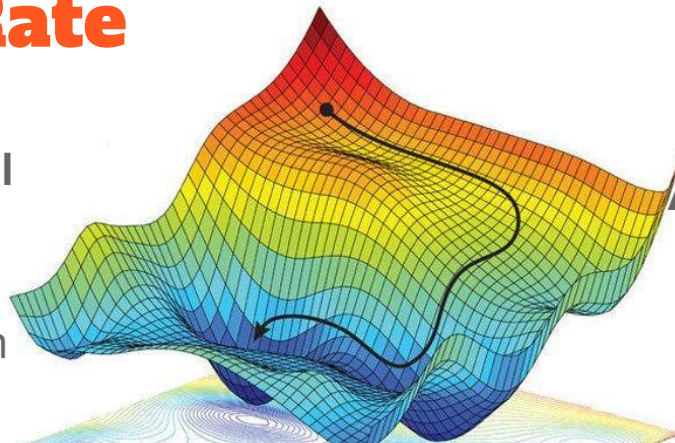
Answer: $f(a+h) - f(a)$ is negative, which means $f(a+h) \leq f(a)$.

Optimizing ANN: Update each ANN weight a as $a - \lambda f'(a)$, where λ is the learning rate.

Importance of Learning Rate

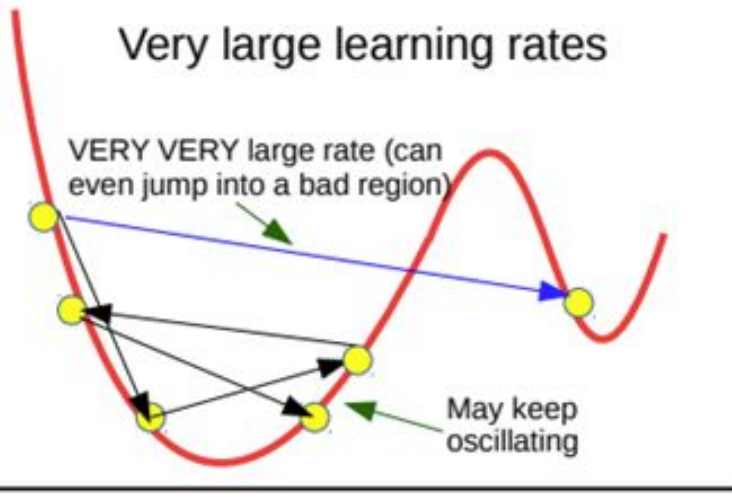


An Ideal
Case
of Loss
function

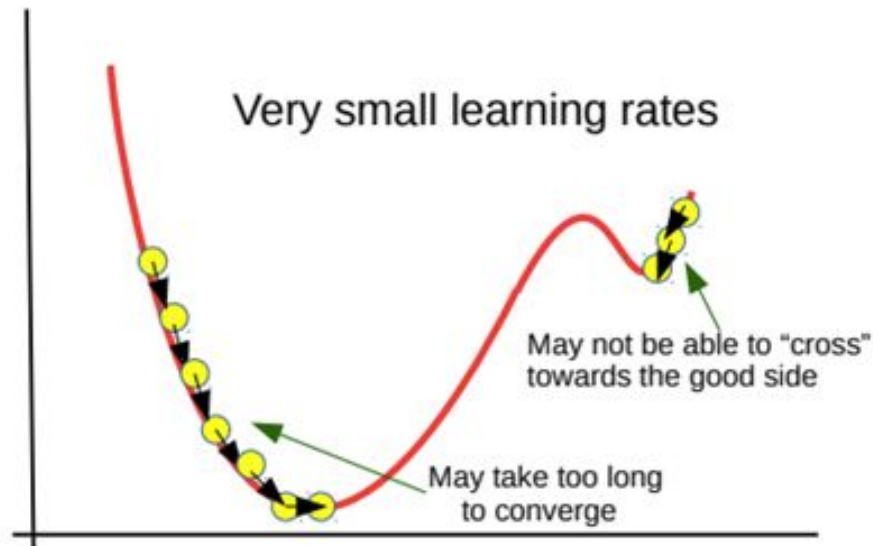


A Practical
Case
of Loss
function

Very large learning rates

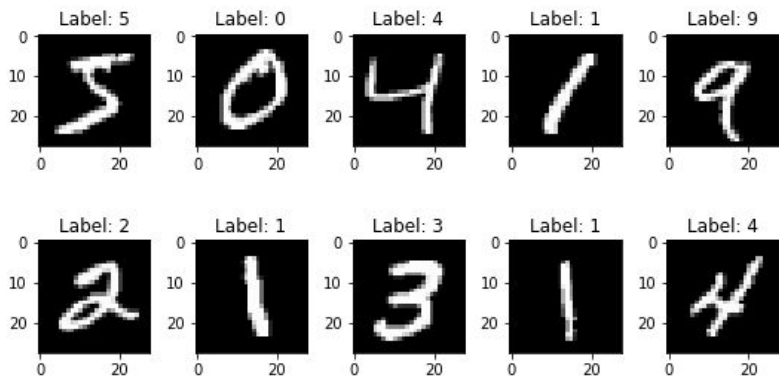


Very small learning rates

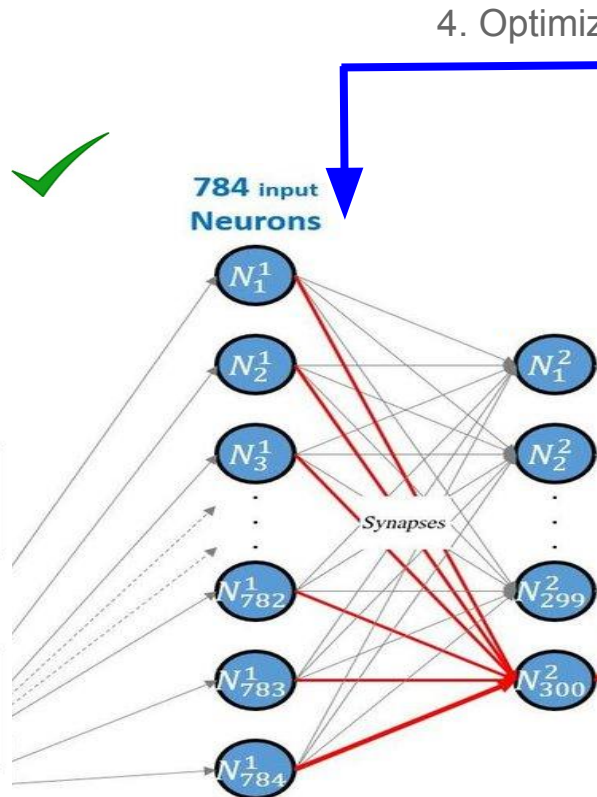


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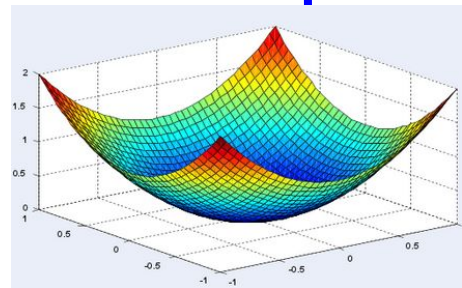
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1. Collect Labelled dataset



2. Design Artificial Neural Network

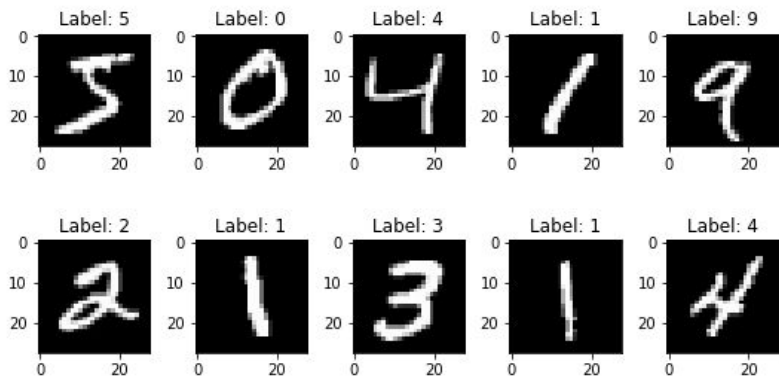


3. Loss Function

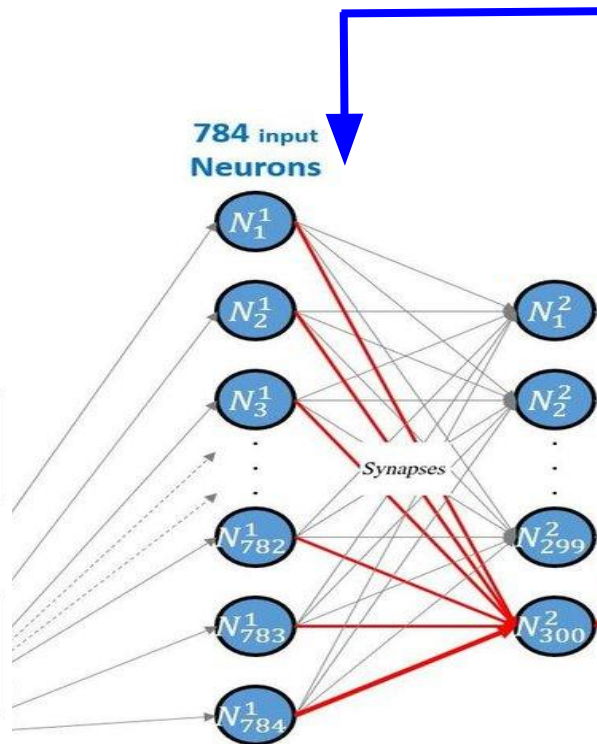
Parametric vs Nonparametric Methods?

4. Optimize Weight

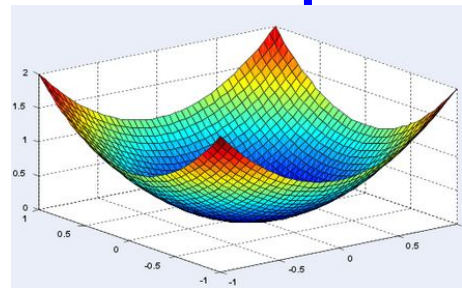
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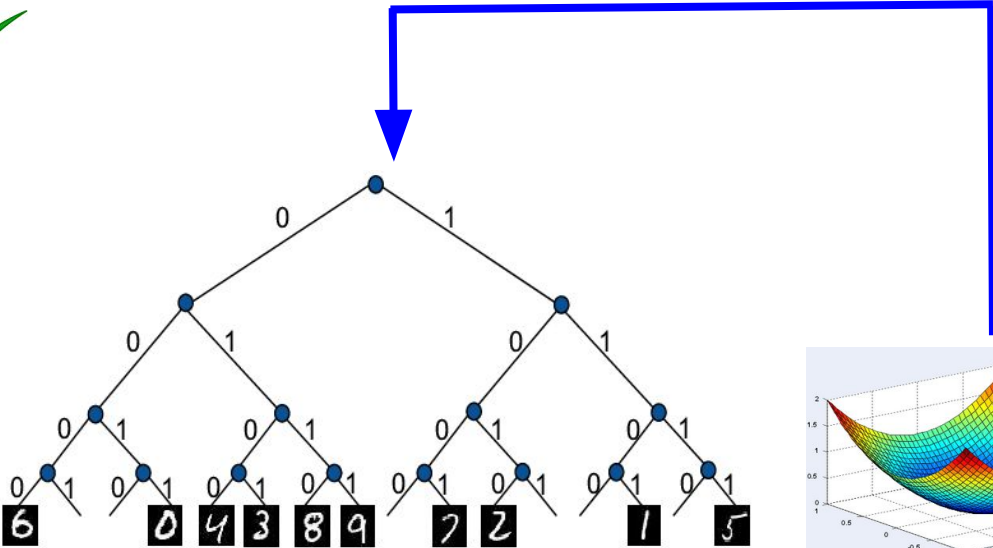
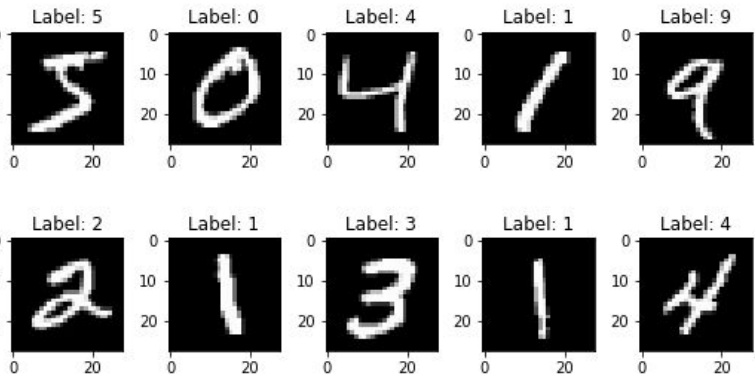


3. Loss Function

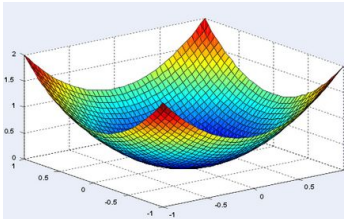
Parametric vs Nonparametric Methods?

4. Optimize Weight

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2. Design Artificial Neural Network



3. Loss Function

1. Collect Labelled dataset



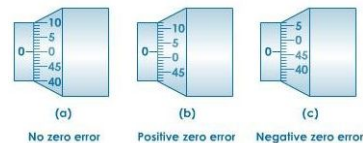
Data Challenge 1: Missing Data

Missing data is a source of error in any data set requiring correction as they can lead to serious problems.

Imputation methods can be used in order to fill those gaps and provide a complete data set.

	School ID	Name	Address	City	Subject	Marks	Rank	Grade
0	101.0	Alice	123 Main St	Los Angeles	Math	85.0	2	B
1	102.0	Bob	456 Oak Ave	New York	English	92.0	1	A
2	103.0	Charlie	789 Pine Ln	Houston	Science	78.0	4	C
3	NaN	David	101 Elm St	Los Angeles	Math	89.0	3	B
4	105.0	Eva	NaN	Miami	History	NaN	8	D
5	106.0	Frank	222 Maple Rd	NaN	Math	95.0	1	A
6	107.0	Grace	444 Cedar Blvd	Houston	Science	80.0	5	C
7	108.0	Henry	555 Birch Dr	New York	English	88.0	3	B

Data Challenge 2: Errors Vs Artifacts



Artifacts



Aspects of
Real World



Raw Data

Worksheet1										Review Sheet		Share					
Home Insert Page Layout Formulas Data Review View														File			
Calibri (Body) 12 A A														General			
B I U L														Conditional Formatting Styles			
Paste														Delete Format			
A1														Format & Filter			
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1		Invertebrates	Amphibians	Reptiles	Birds	Mammals	Fish	Insects	Plants	Fungi	Protists	Other					
2		100	97.5	95.5	94.1	88.2	71.2	71.1	74.3	74.5	71	75.6	73.8	75.1	72.8	67.8	66.5
3		98.7	100	95.4	93.9	87.9	71.2	71	74.4	74.5	75.2	75.8	73.8	75.3	72.7	67.8	66.7
4		97.4	97.4	100	94.2	88.1	71.3	71.1	74.4	74.5	75.4	75.7	74	75.2	72.6	67.7	66.6
5		96.6	96.5	96.5	100	87.9	70.8	70.5	74.2	74.3	75.9	75.5	73.6	74.8	72.6	67.3	66.6
6		95	93	92.9	100	79.6	70.4	73.6	73.6	73.6	74.4	74.4	72.6	74.5	72	67	66.2
7		93.9	94	94	92.7	83.4	100	97.8	74.5	75	75.6	75.4	73.8	76.7	74.4	69.8	68.7
8		93.9	94	93.8	93.5	83.3	93.4	100	74.7	74.7	75.5	75.8	73.8	76.5	74.4	69.8	68.5
9		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
10		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
11		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
12		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
13		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
14		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
15		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
16		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
17		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
18		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
19		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
20		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
21		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
22		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
23		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
24		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
25		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
26		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
27		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
28		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
29		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
30		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
31		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
32		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
33		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
34		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
35		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
36		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
37		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
38		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
39		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
40		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
41		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
42		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
43		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
44		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
45		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
46		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
47		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
48		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
49		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
50		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
51		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
52		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
53		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
54		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
55		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
56		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
57		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
58		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
59		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
60		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
61		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
62		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
63		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
64		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
65		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
66		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
67		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
68		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
69		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
70		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
71		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
72		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
73		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
74		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
75		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
76		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5	73.8	76.1	75.9	70.9	71.6
77		93.9	93.9	93.9	93.8	83.3	87.1	88.9	74.9	74.9	75.9	76.5					

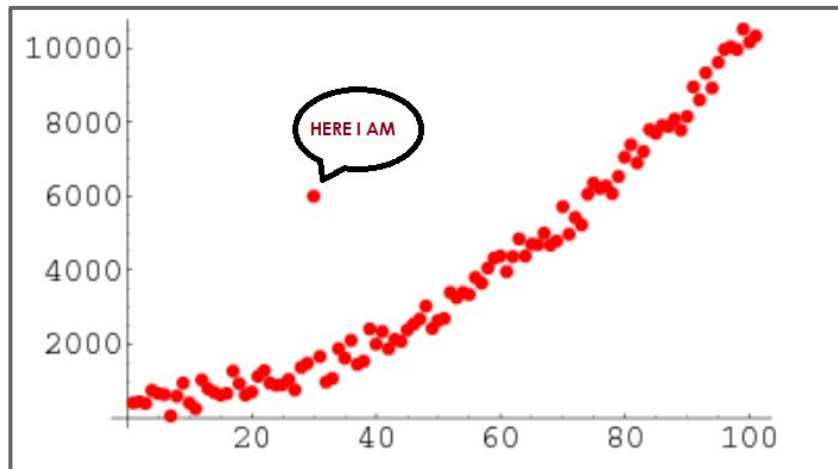
Processed Data

Data Challenge 3: Outliers

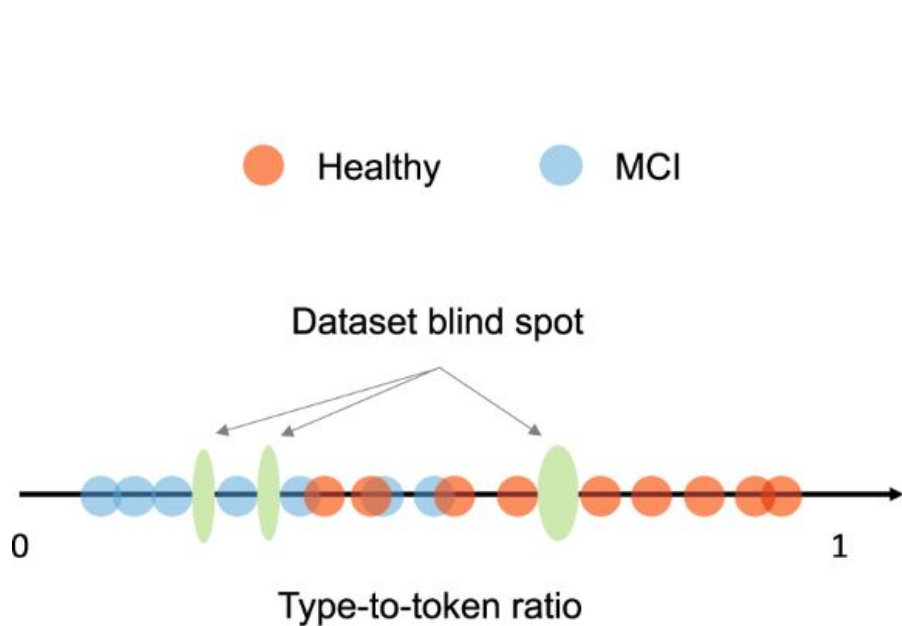
Outlier: A data object that **deviates significantly** from the normal objects as if it were **generated by a different mechanism**

Ex.: Unusual credit card purchase

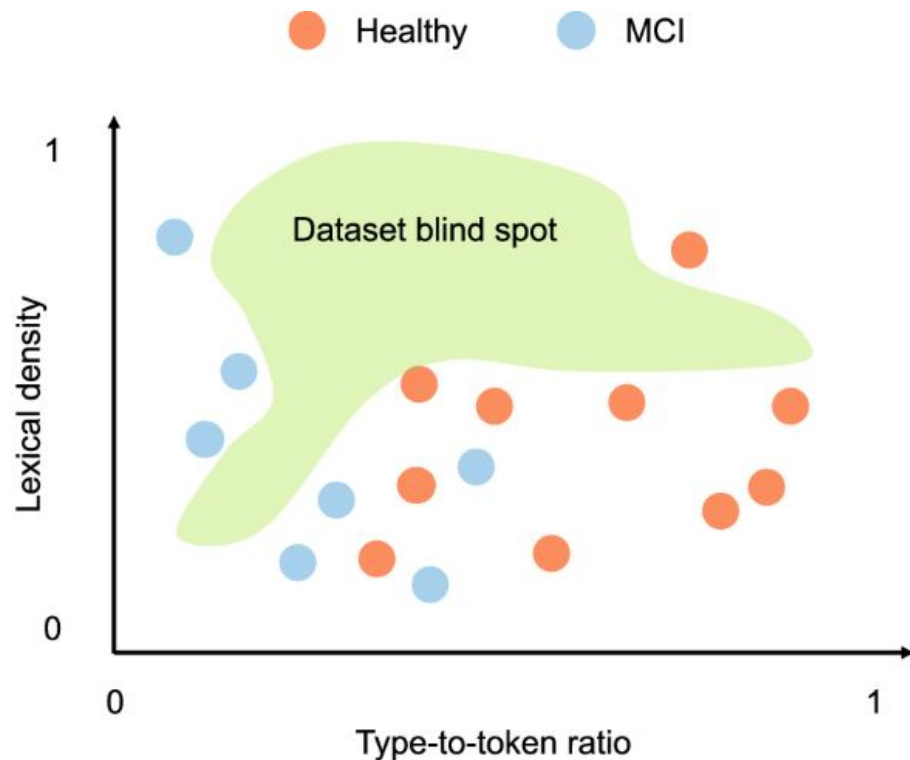
- Outliers are different from the Errors
- Outliers are interesting: It violates the mechanism that generates the normal data



Data Challenge 4: Curse of Dimensionality



(a)



(b)

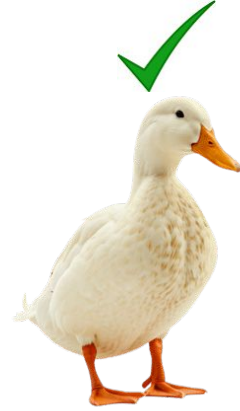
More Data Requirement

Data Challenge 4: Curse of Dimensionality



Requirement of Complex ANN Model

Model Challenge 2: Inductive Bias



Model Challenge 3: Evaluation

		Reality	
		True	False
Measured or Perceived	True	Correct 	Type 1 error False Positive
	False	Type 2 error False Negative	Correct 

Error Types: Case Study



Cancer Diagnostics



Anti-missile System



Weather Predictions

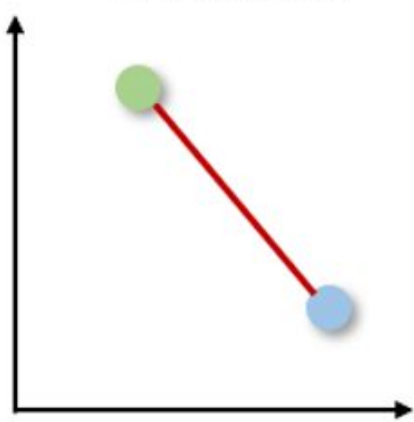
Performance Metrics

Key: **tp** = True Positive, **tn** = True Negative, **fp** = False Positive, **fn** = False Negative

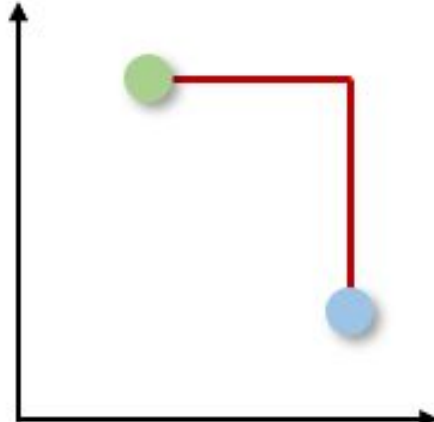
Metric Name	Metric Formula	When to use
Accuracy	Accuracy = $\frac{tp + tn}{tp + tn + fp + fn}$	Default metric for classification problems. Not the best for imbalanced classes.
Precision	Precision = $\frac{tp}{tp + fp}$	Higher precision leads to less false positives.
Recall	Recall = $\frac{tp}{tp + fn}$	Higher recall leads to less false negatives.
F1-score	F1-score = $2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$	Combination of precision and recall, usually a good overall metric for a classification model.
Confusion matrix	NA	When comparing predictions to truth labels to see where model gets confused. Can be hard to use with large numbers of classes.

Model Challenge 4: Loss or Scoring Function

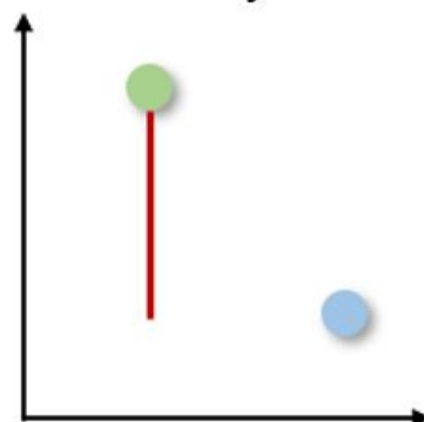
Euclidean



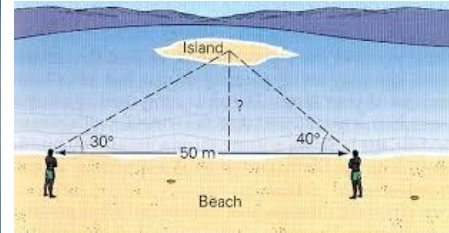
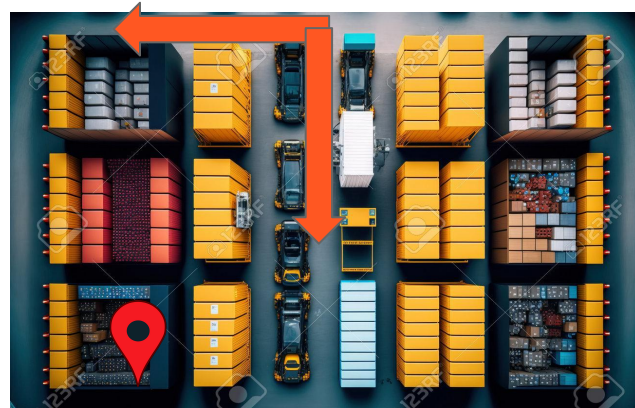
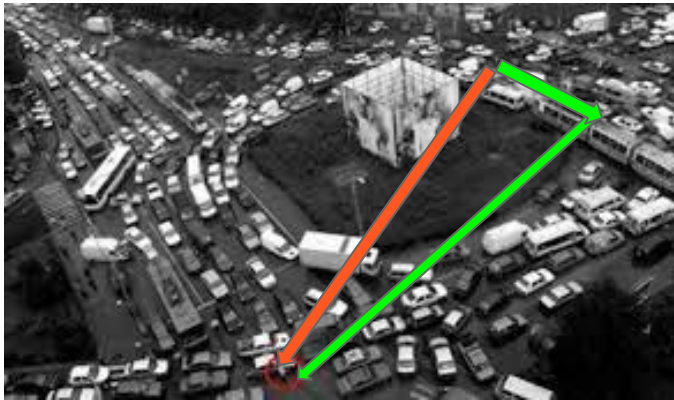
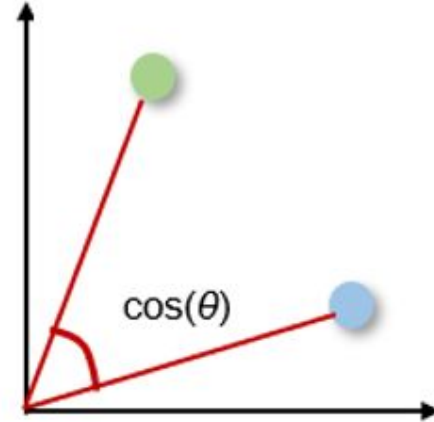
Manhattan



Chebyshev



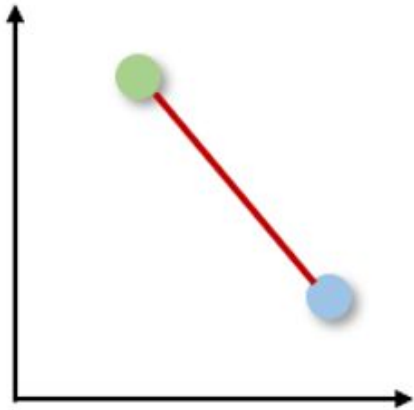
Cosine



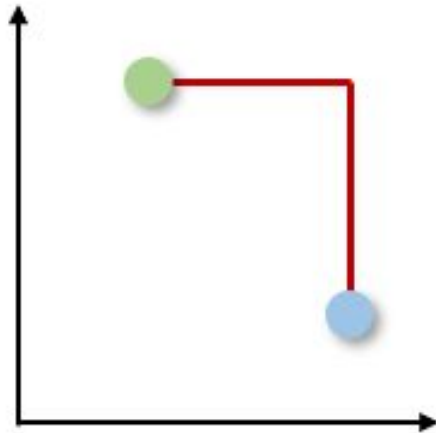
Model Challenge 4: Loss or Scoring Function?



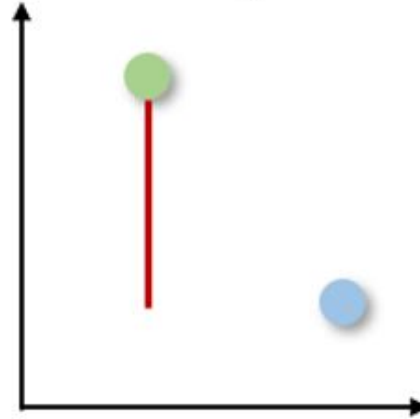
Euclidean



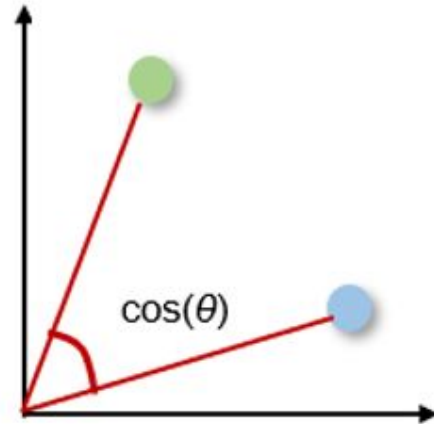
Manhattan



Chebyshev



Cosine



Jumbled?

$$d = \max_i (|x_i - y_i|)$$

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

$$d = \frac{\langle x y \rangle}{\|x\| \|y\|}$$

$$d = \sum_{i=1}^n |(x_i - y_i)|$$

Introduction to Unsupervised Learning

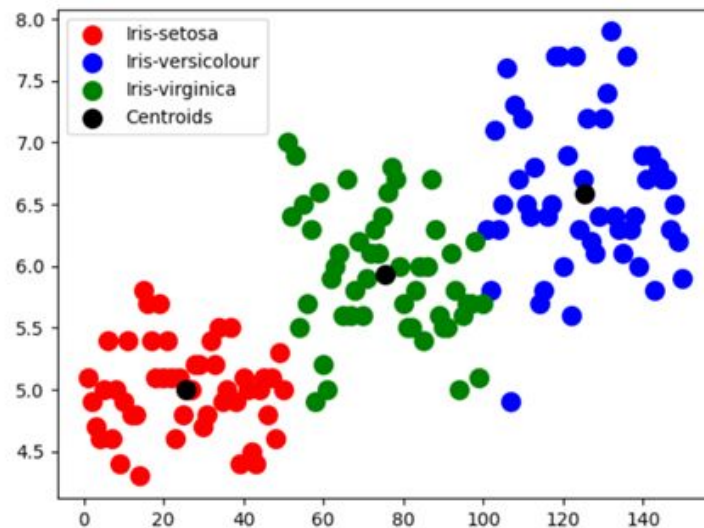
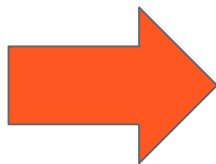
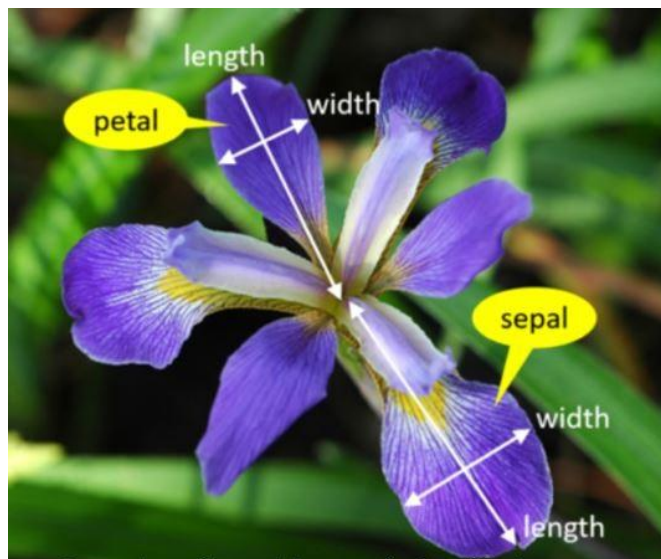




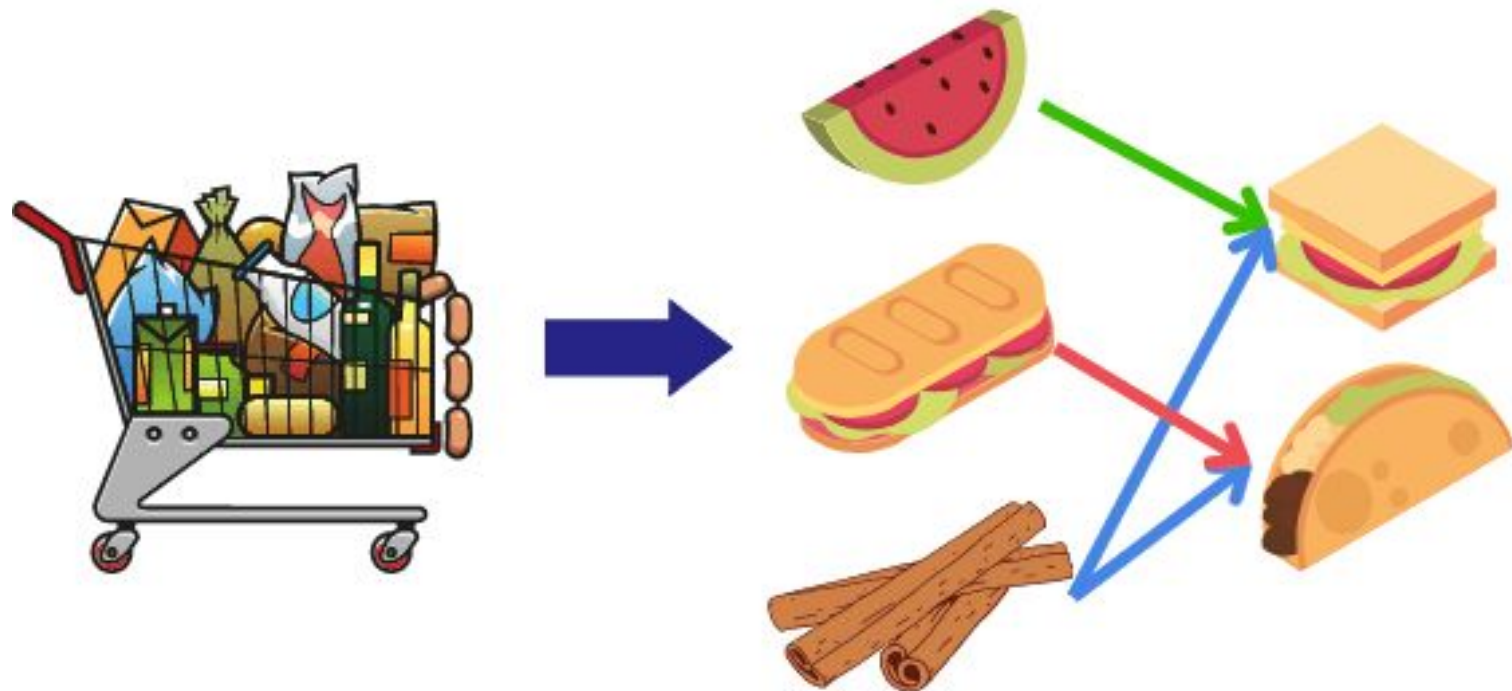
VS



Clustering



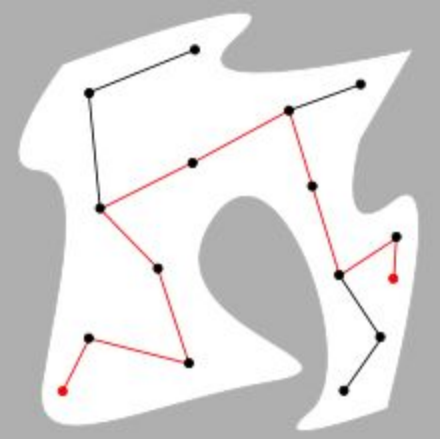
Association



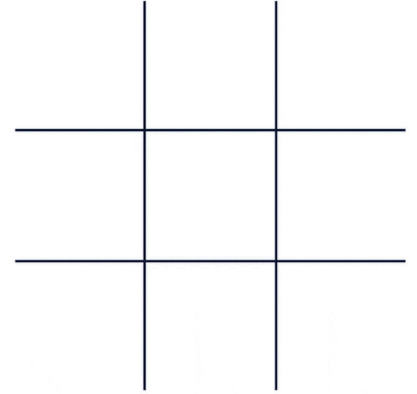
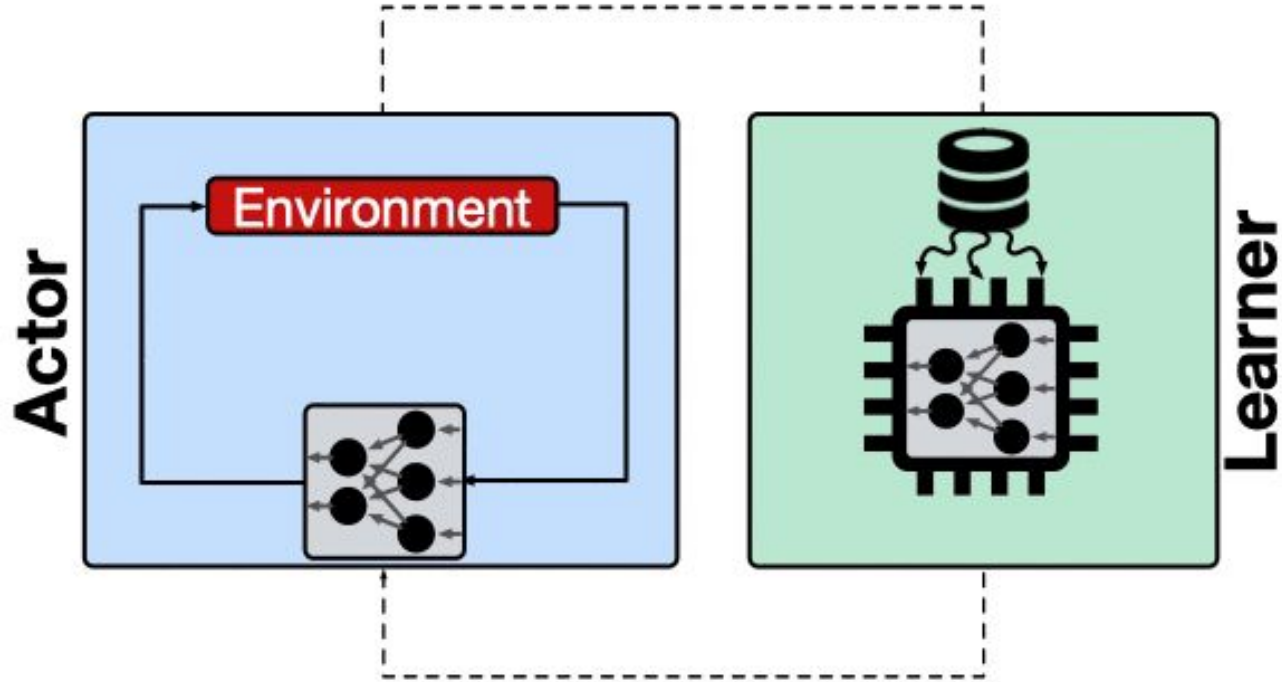
*"93% of people who purchased item A
also purchased item B"*

Introduction to Reinforcement Learning



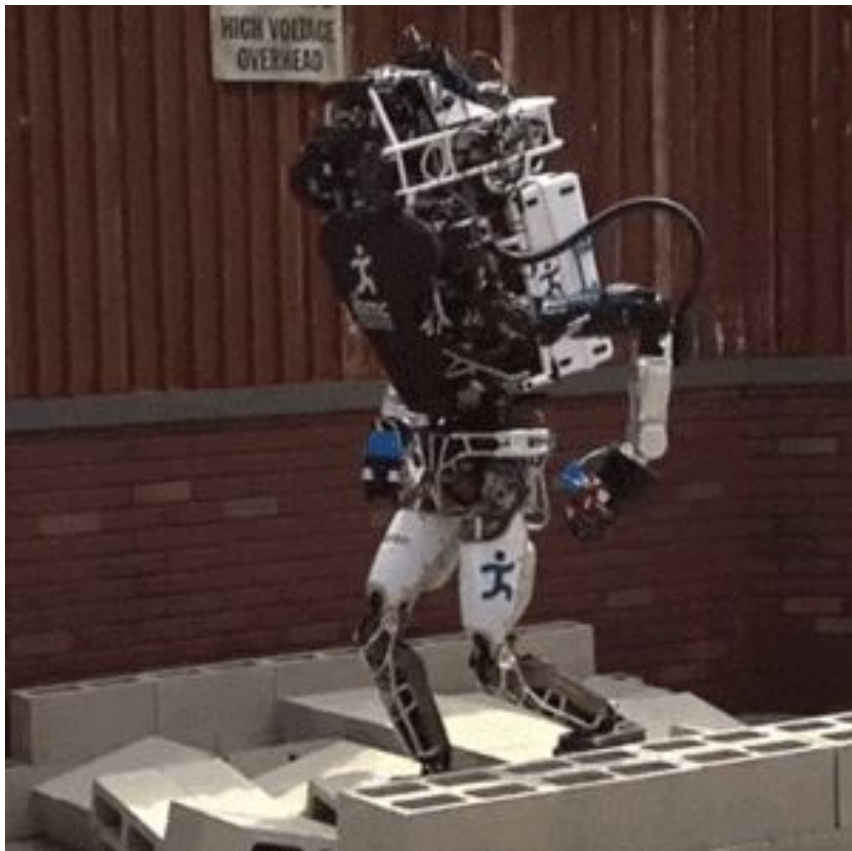


Reinforcement Learning



Reinforcement Learning

Objective: Move forward while minimizing how often it fails and the energy it expends.



Learning Phase

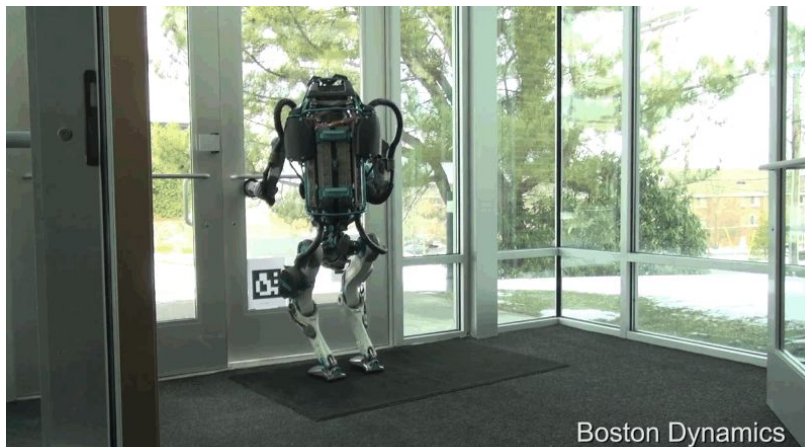
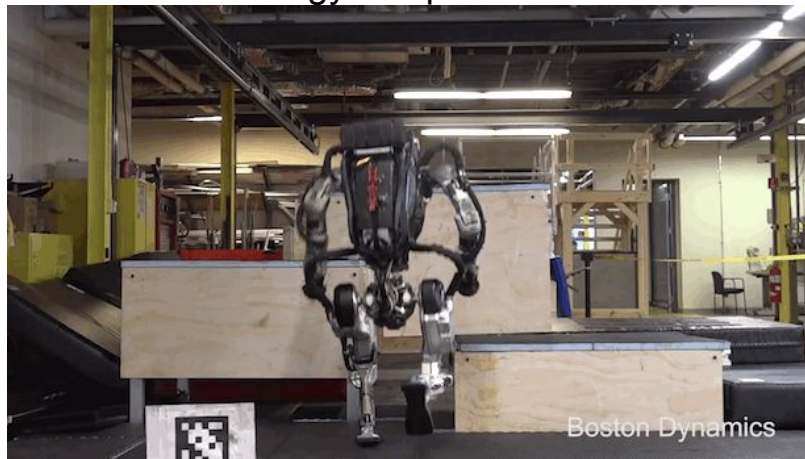


Reinforcement Learning

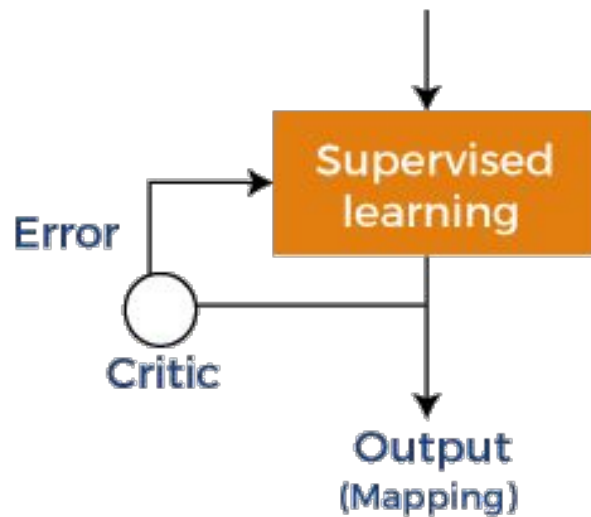
Objective: Move forward while minimizing how often it fails and the energy it expends.



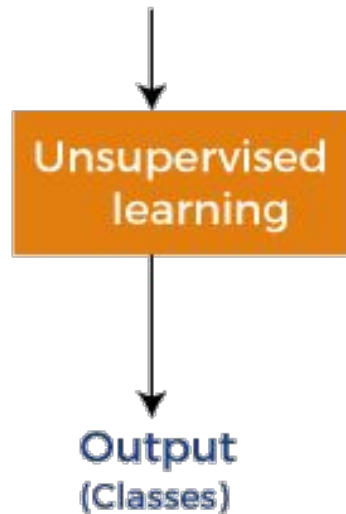
Learning Phase



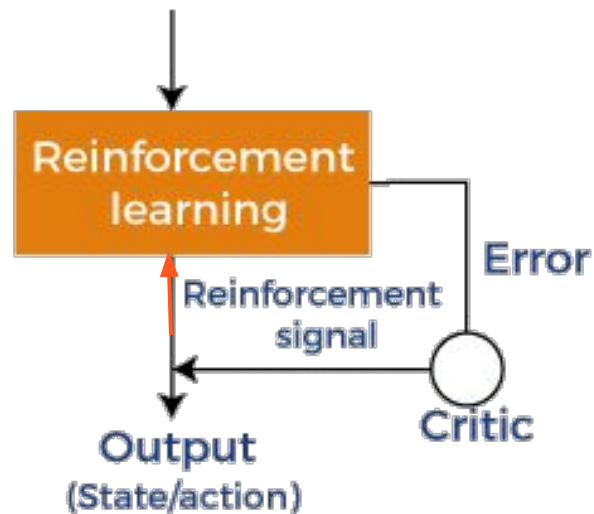
(Data with labels)
Input



(Data without labels)
Input



(States and actions)
Input



Summary of “Intro to Machine Learning”

